

HDR Testing 101

Code == 07-01-25

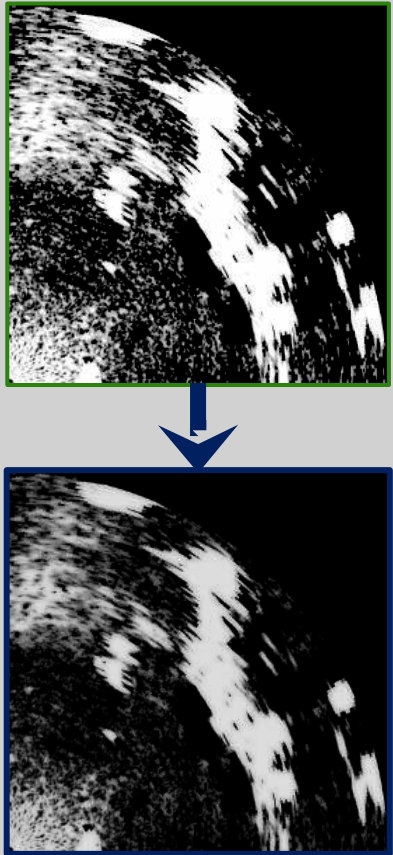
A summary

– J.C. Vaught

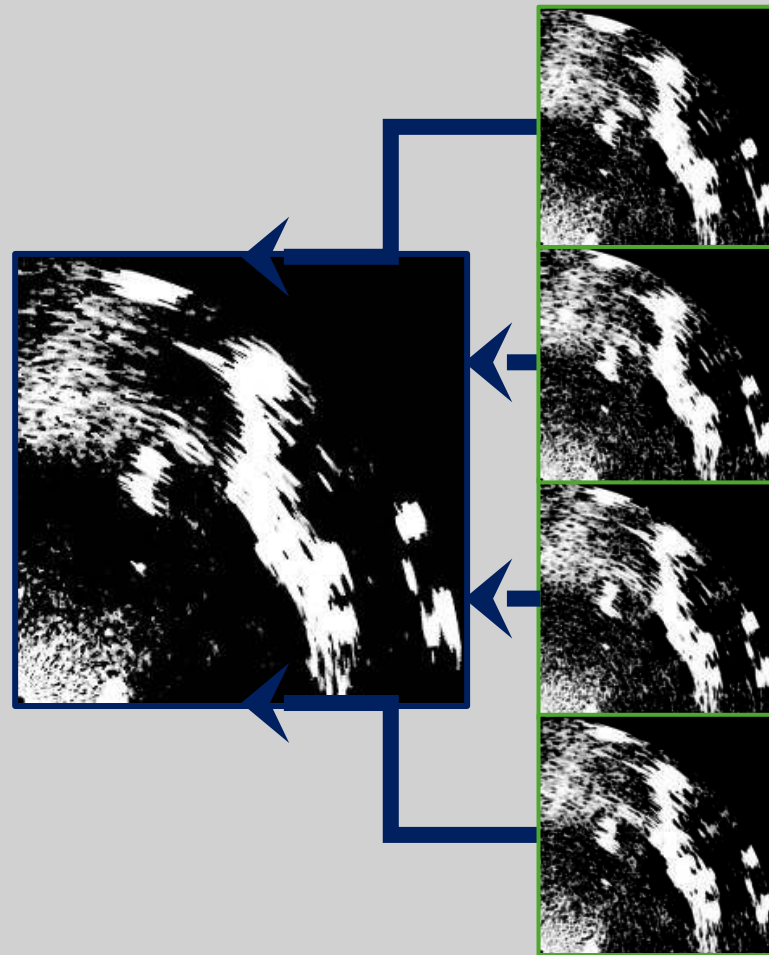


PART 1: The 3 Algorithm Classes

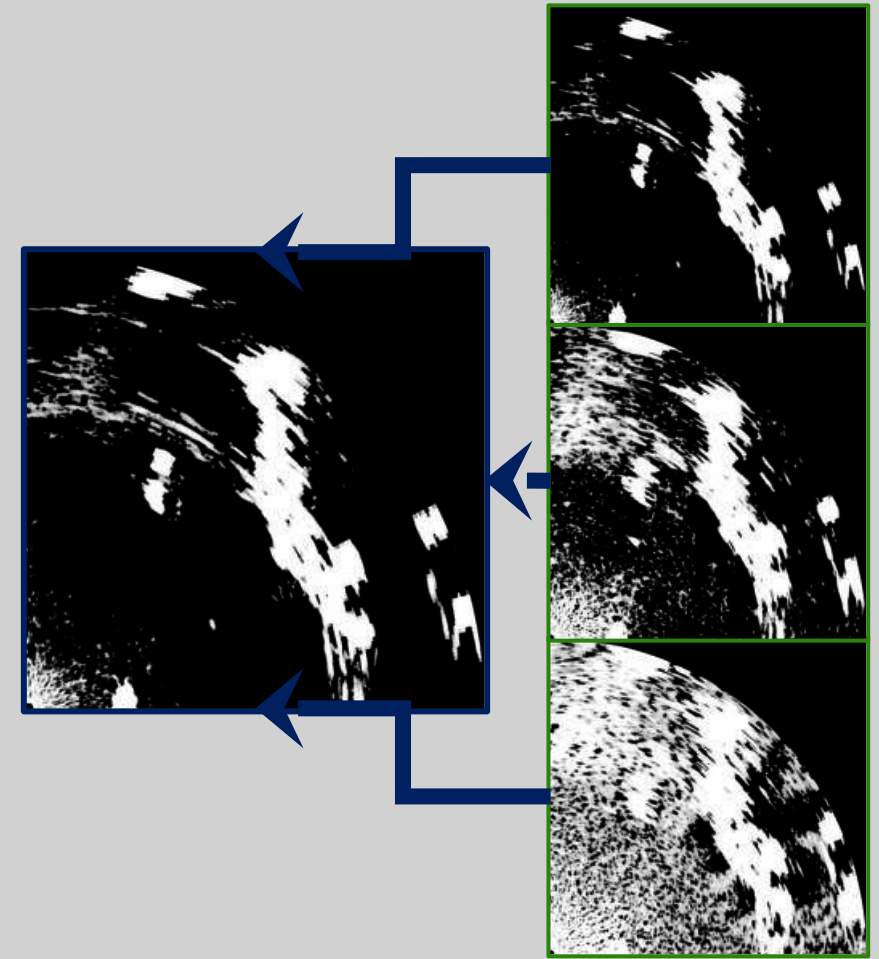
Single-Frame noise reduction models



Multi-Frame noise reduction models



Multi-Exposure noise reduction models





Single-Frame Algorithms

Methods for clearing noise/clutter with single frame algorithms

Camera and ML based techniques

Lee Filter, Noise2Void, BM3D, Wavelets

Lee Filtering

- **How It Works**

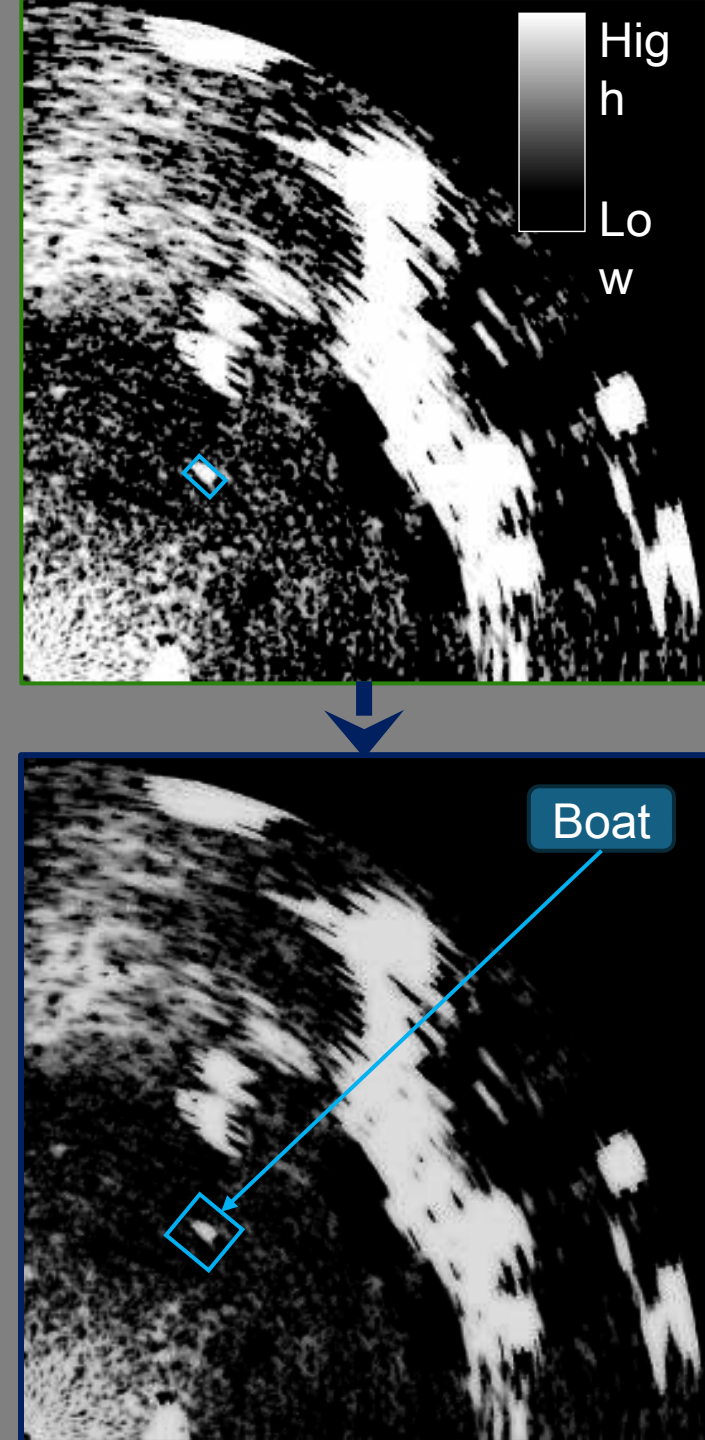
- Uses a local window to compute mean & variance
- Adapts smoothing based on local variance (less smoothing at edges, more in flat regions)
- Preserves features better than a simple blur

- **Best For**

- Reducing LIGHT speckle
- Smoothing large uniform areas
- Preserving main edges/targets

- **Limitations**

- Can blur small or thin features
- Strong edges may still get softened



Noise2Void – Self Supervised

- **Core idea:**

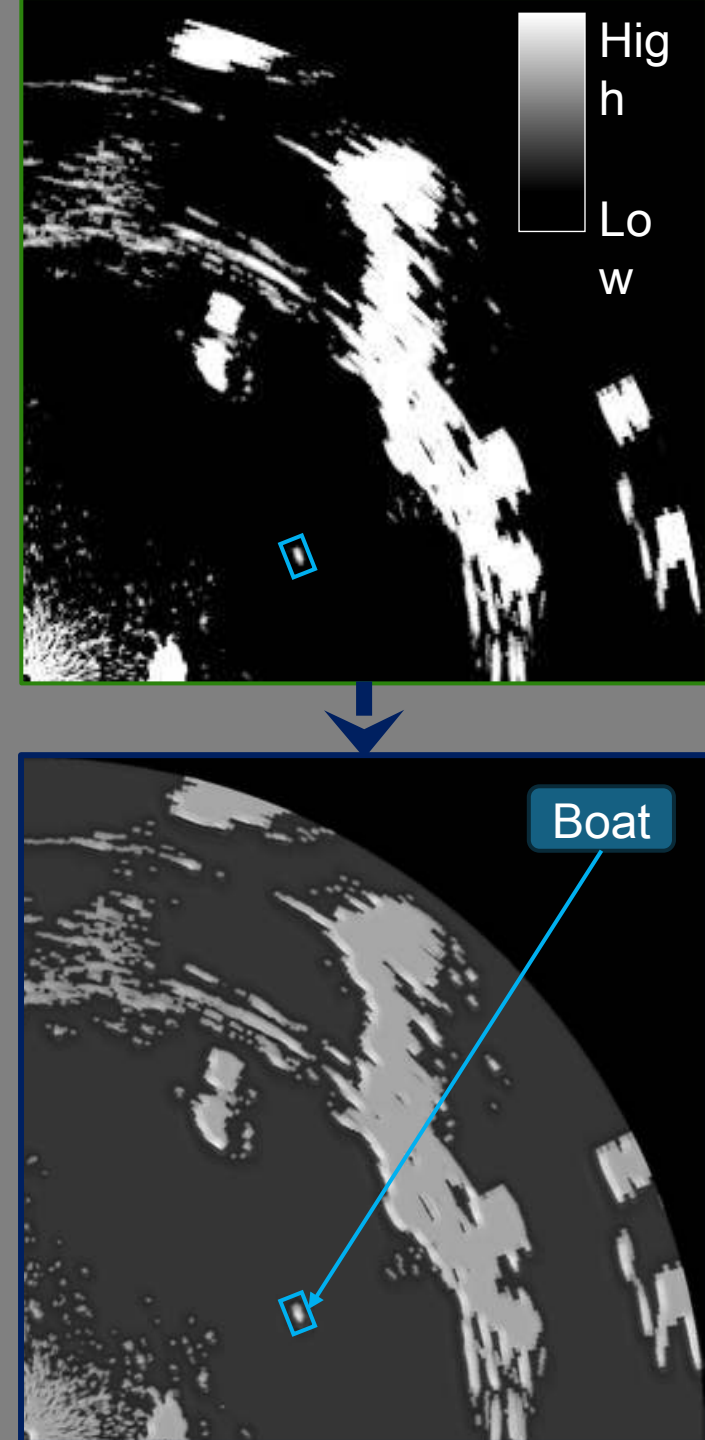
- Train on *noisy-only* images
- Randomly mask ($\sim 1\text{--}5\%$) “blind-spots” in each patch
- U-Net predicts those pixels using surrounding context
- Inference: feed full noisy image $\rightarrow$ denoised output

- **Strengths:**

- Adapts to complex, unknown noise (speckle, Poisson, mixed)
- No clean ground-truth required; ideal for real-world radar data
- Preserves edges & fine details via multi-scale context

- **Limitations:**

- Fails on large-scale/structured artifacts (stripes, bands)
- Smooths out truly unique/rare features (isolated point targets)



BM3D Denoising

- **How it works:**

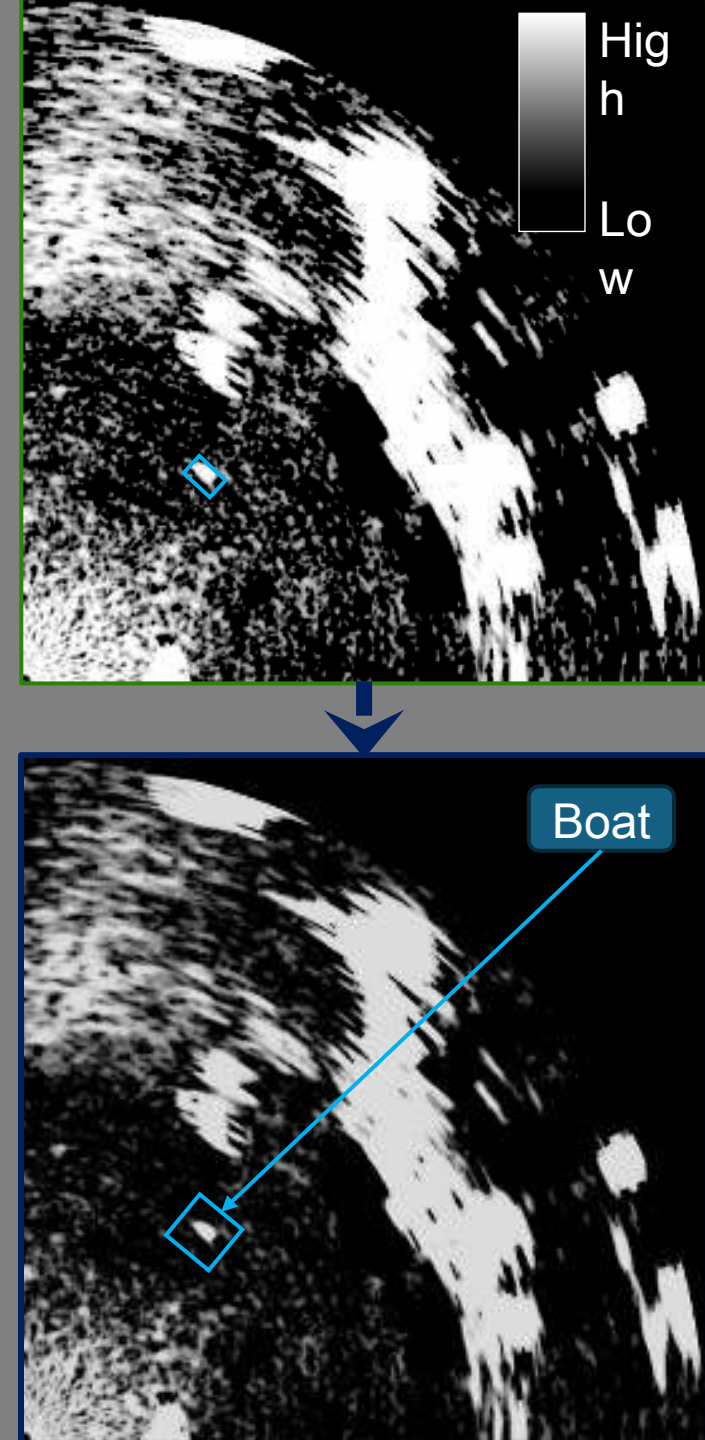
- Finds similar patches across the image
- Groups and filters them in 3D to suppress noise
- Reconstructs a cleaner image, preserving edges

- **Works best when:**

- Noise is random (“grainy”) and not too strong
- True signal is present but slightly obscured
- Fine details and textures need preservation

- **Works poorly :**

- Most pixels are just min or max (saturated)
- Image is heavily quantized or “salt & pepper” noisy
- Information is lost due to clipping or extreme sparsity



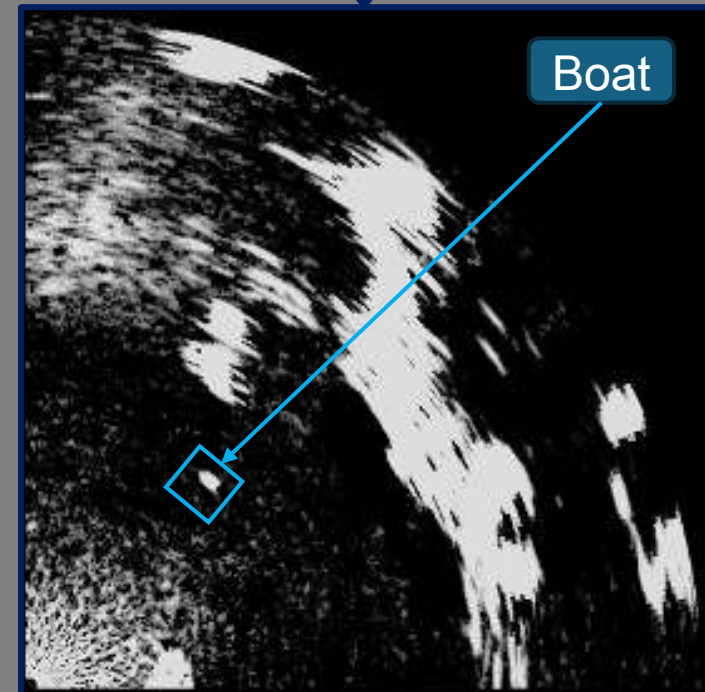
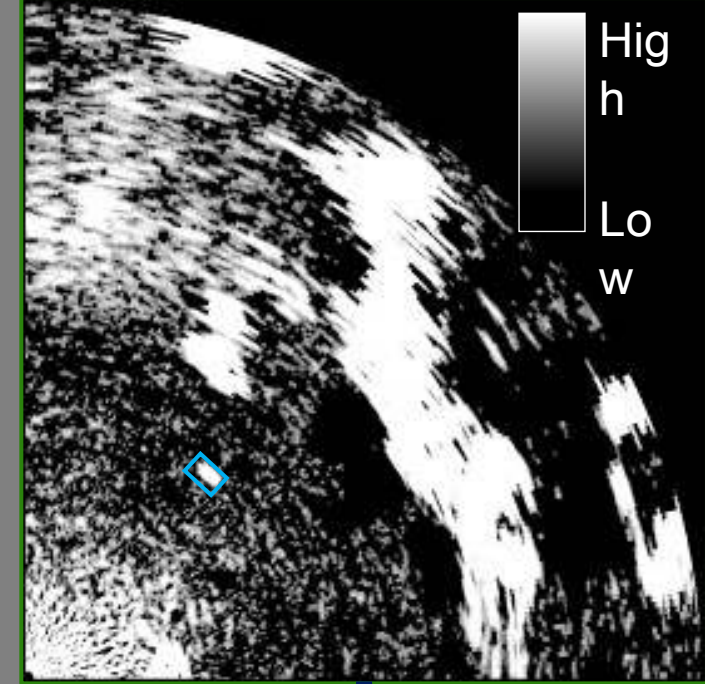
Wavelet Decomposition

- **Strengths:**

- **Reduces speckle/clutter:** Cleans up noisy backgrounds while keeping large, real targets.
- **Edge-preserving:** Better at keeping sharp features than basic smoothing or blurring.
- **Multi-scale:** Removes noise at multiple levels of detail at once.

- **Downfalls:**

- **Erase small/weak targets:** Faint signals (e.g., buoy at low gain) may be lost as “noise”
- **Artifacts:** May introduce blocky patterns, ringing, or strange edges, especially at high denoising strength or with certain wavelets.
- **Tuning needed:** Choice of wavelet, threshold, and parameters has a big impact





Single Gain Methodologies

Methods for clearing noise/clutter from Radar Data at a constant Gain.

Primarily Camera-based techniques, but with some Radar specific techniques as well

MTI, RPCA, NLM, & Multi-Look Averaging

MTI (Radar Specific)

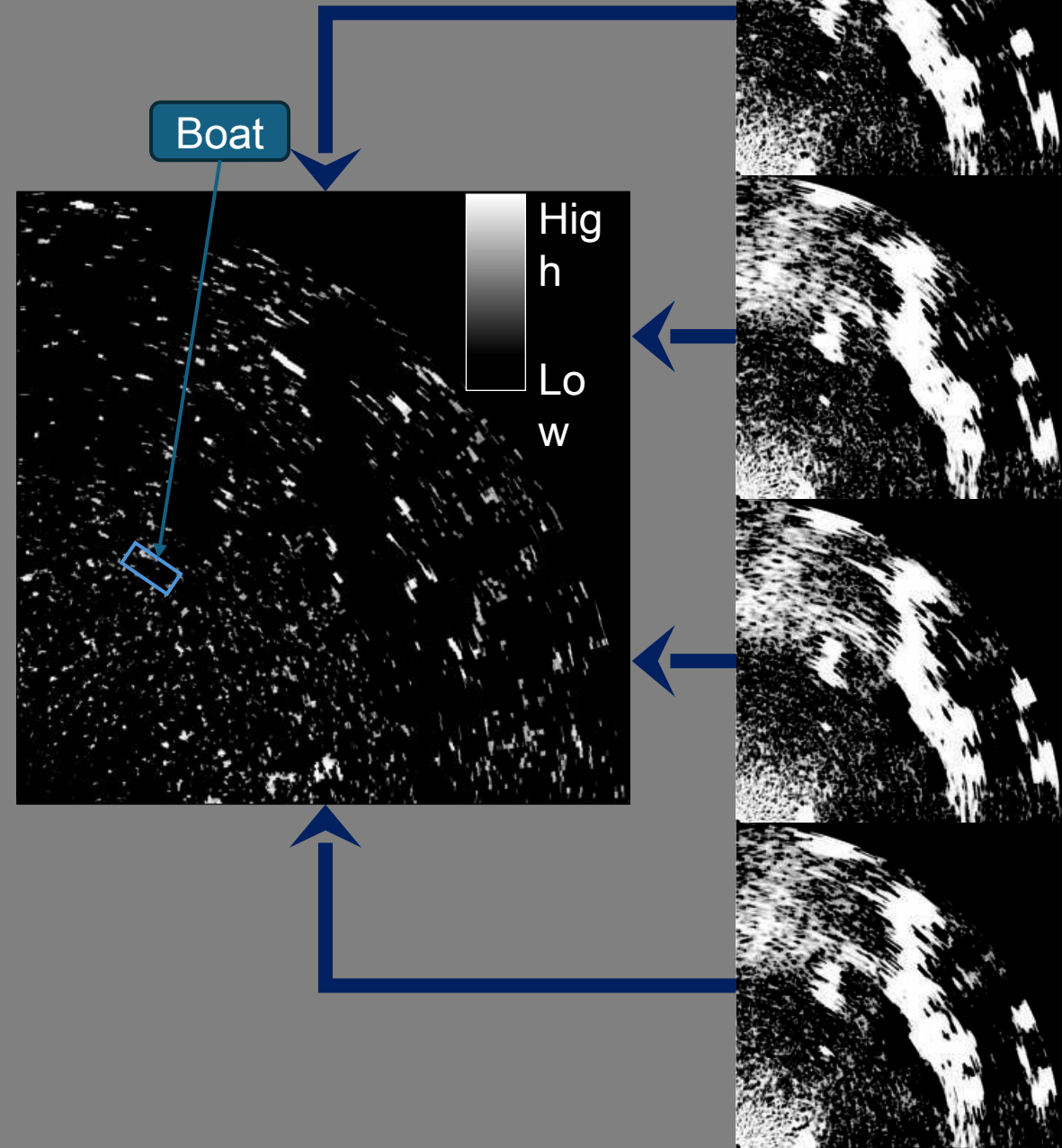
a. What MTI is supposed to do:

- a. MTI (Moving Target Indication) tries to remove persistent echoes, ideally leaving only moving targets

b. What goes wrong here?

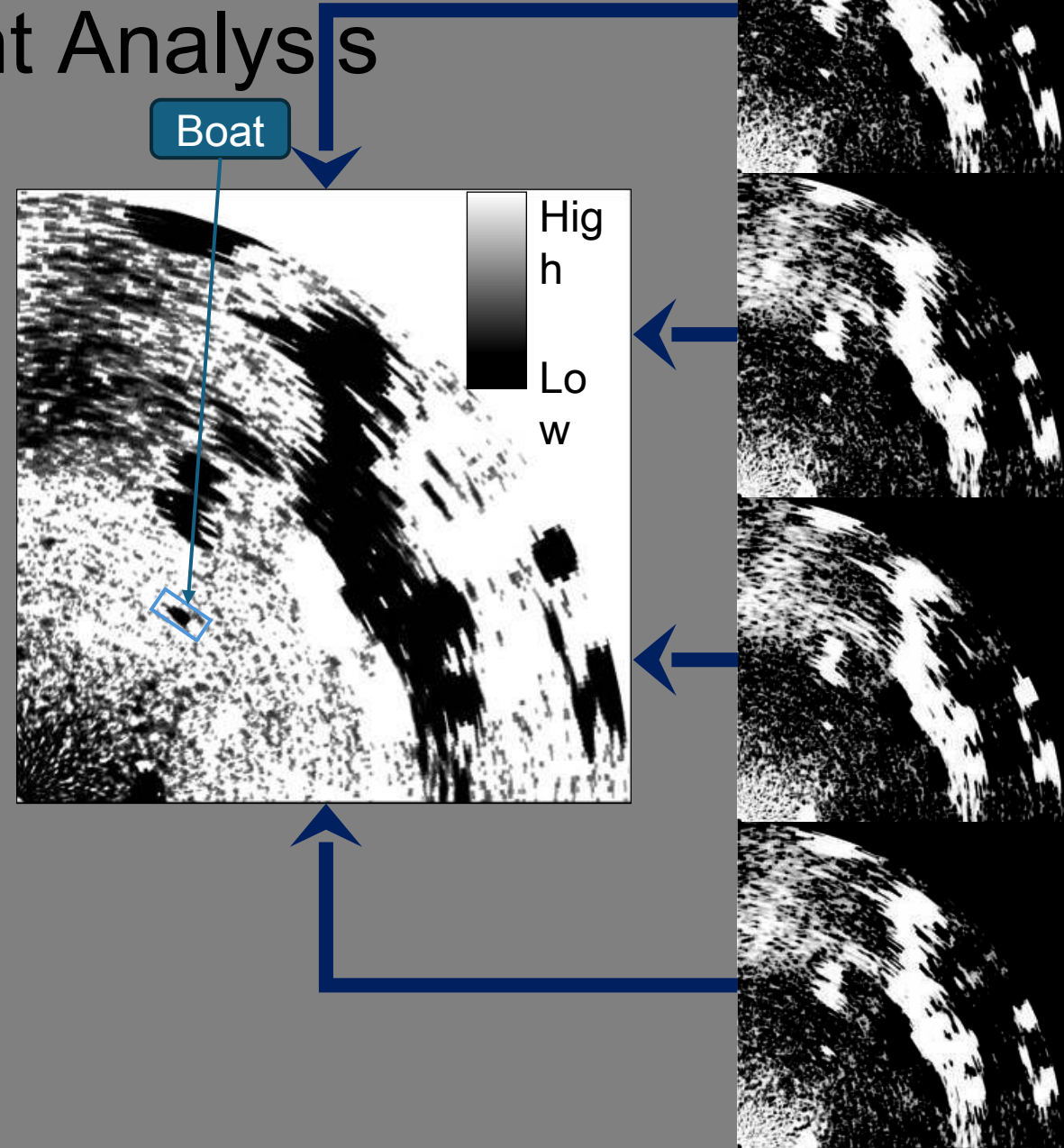
- a. In real marine radar, the background (sea, clutter) often *isn't stable*. The radar may move, vibrate, or scan at slightly different angles ('azimuth shift')
- b. Every time the background shifts, MTI thinks all those changes are 'new targets'

c. Output of median-based MTI subtraction on Radar data during thunderstorm -->



Robust Principal Component Analysis

- **RPCA Splits radar frames**
 - **Background (Low-rank):** Persistent clutter (sea, land, stationary targets)
 - **Foreground (Sparse):** Transient, moving, or unusual returns (boats, birds, rain)
- **RPCA is useless... kinda**
 - Stationary objects (buoys, anchored boats) get absorbed into the background
 - Slow-moving objects (drifting targets) may also be treated as background
 - RPCA only highlights changes. Anything present in all frames is *suppressed*



Multi-Look Averaging_(or median)

- **How:**

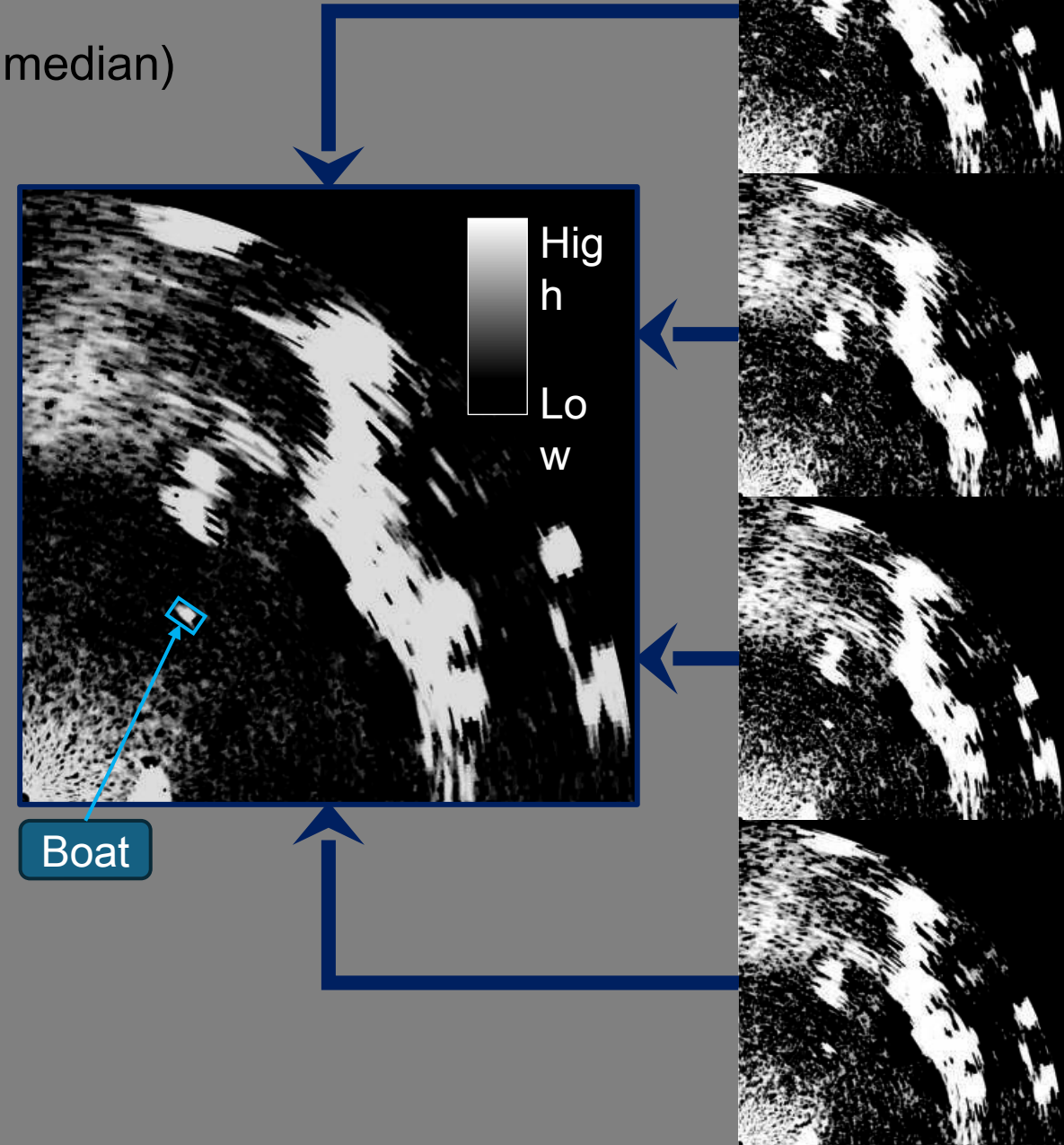
- Combines multiple radar frames of the same scene
- Uses pixel-wise median/mean to reduce speckle noise
- Produces a cleaner, more stable radar image

- **Best at:**

- Stationary or slowly changing features
- Background clutter and persistent targets

- **Limitations:**

- Moving targets may blur or fade
- Less effective with rapid scene changes



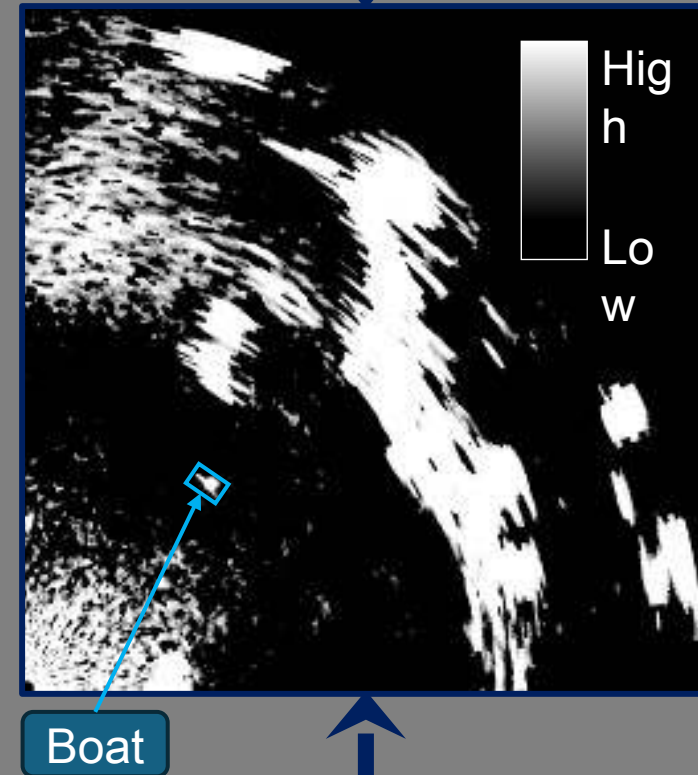
Non-Local Means Denoising

- **Strengths of Multi-Frame NLM**

- Greatly reduces random background noise
- Preserves slow-moving or stationary targets
- Keeps fine detail and structure sharp
- Enhances signal-to-noise ratio for weak targets
- Works especially well in high clutter conditions

- **Some Downsides**

- Fast-moving objects may be attenuated or lost
- Faint ghosting or smearing
- Detail loss if parameters are too aggressive





Multi-Gain Algorithms

Methods for clearing noise/clutter with multiple amplification values

Camera and ML based techniques

NLM, MergeMertens(OpenCV)... **[TODO add more items]**

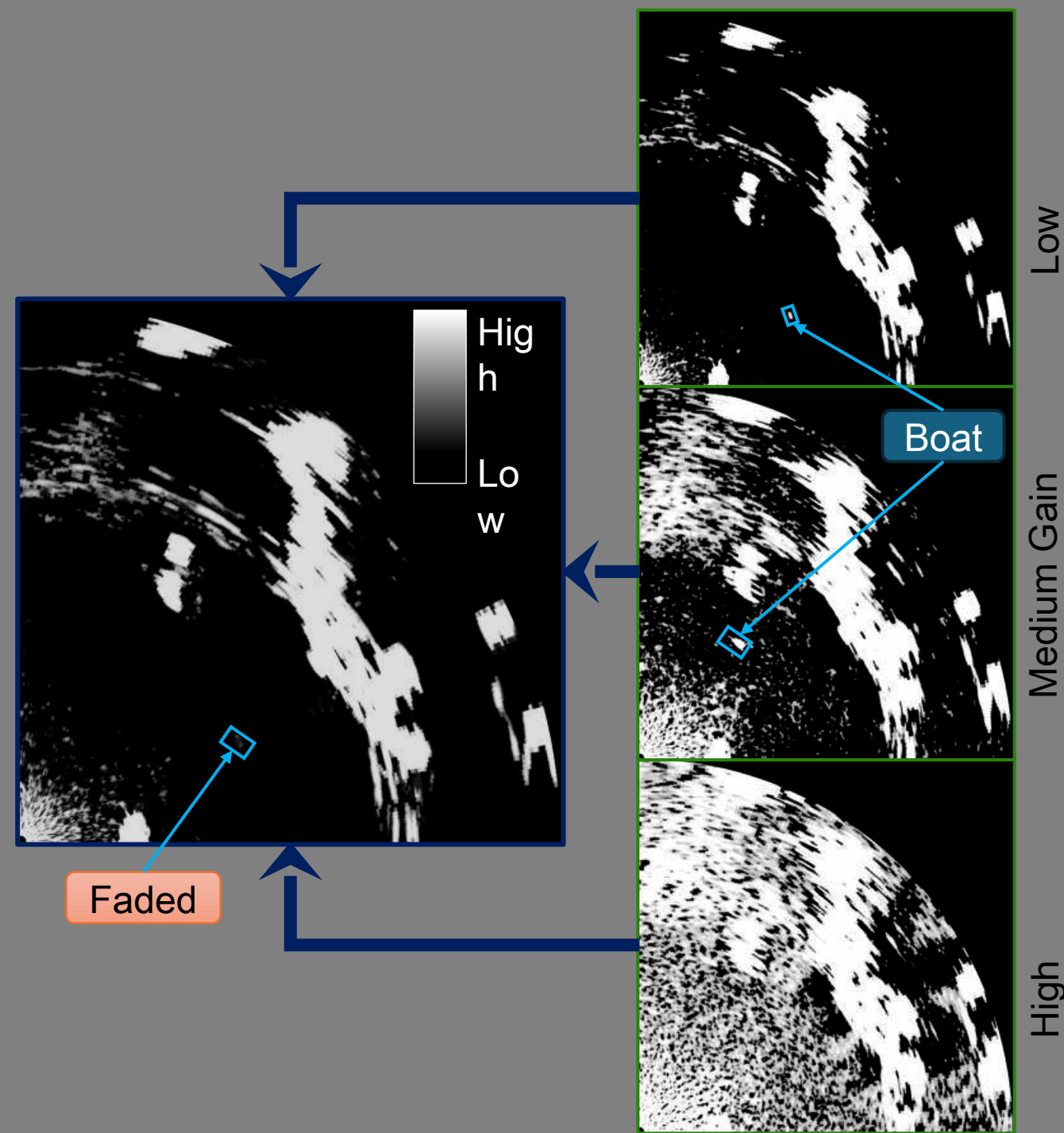
Multi-Gain NLM

- **NLM + Differing Gain**

- Multiple radar sweeps are captured at different gain settings
- Images are stacked and denoised together
- Real targets stay consistent; noise varies with gain and gets suppressed

- **Results**

- **Cleaner background:** Most random noise reduced (better than constant NLM)
- **Faint targets:** Better visibility compared to single-gain images
- **Some clutter/artifacts:** Strong, persistent clutter remain and moving objects faded out



HDR via MergeMerten (OpenCV)

- **Exposure Fusion**

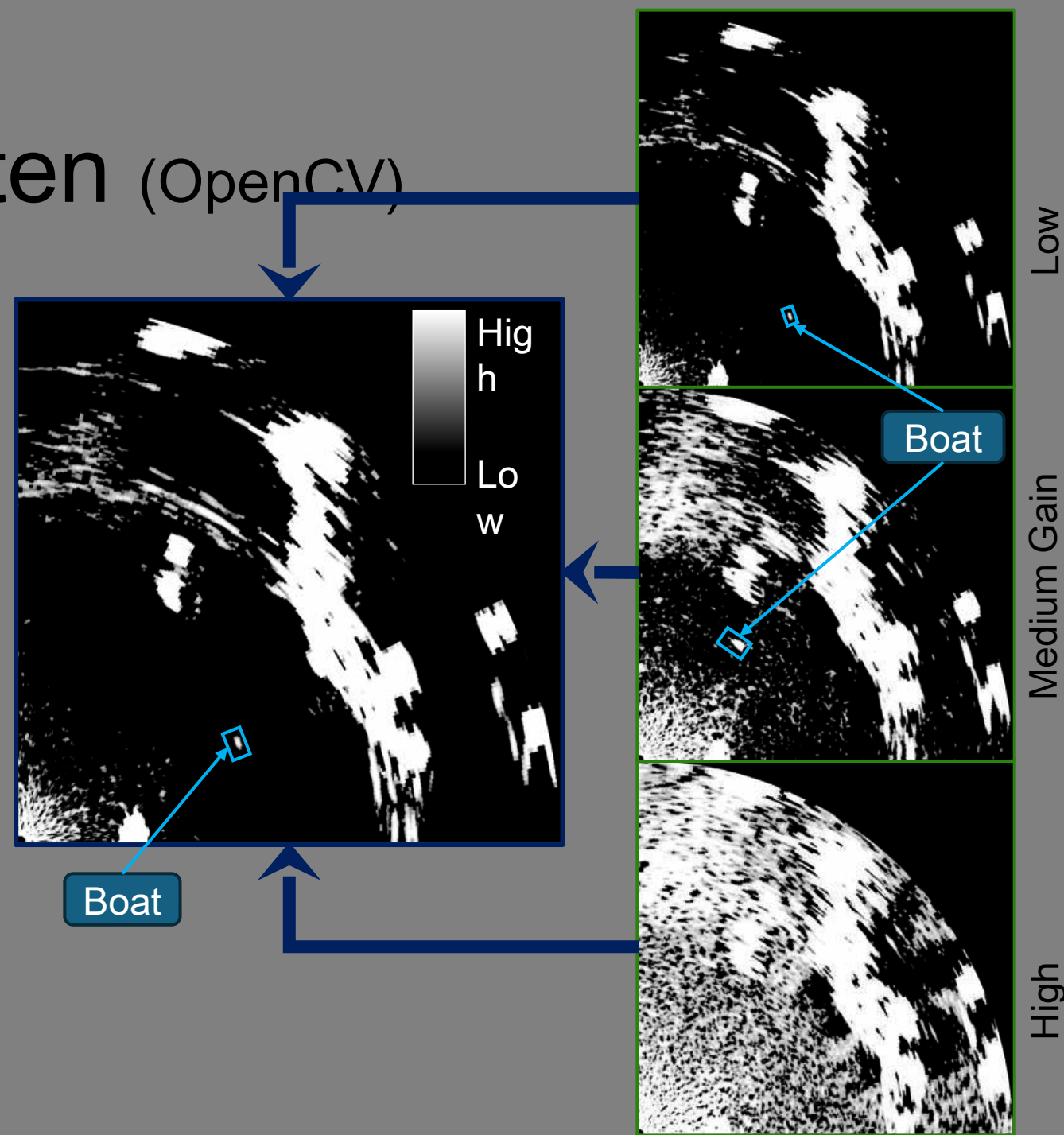
- The algorithm automatically picks out the best parts (well-exposed regions) from each input.

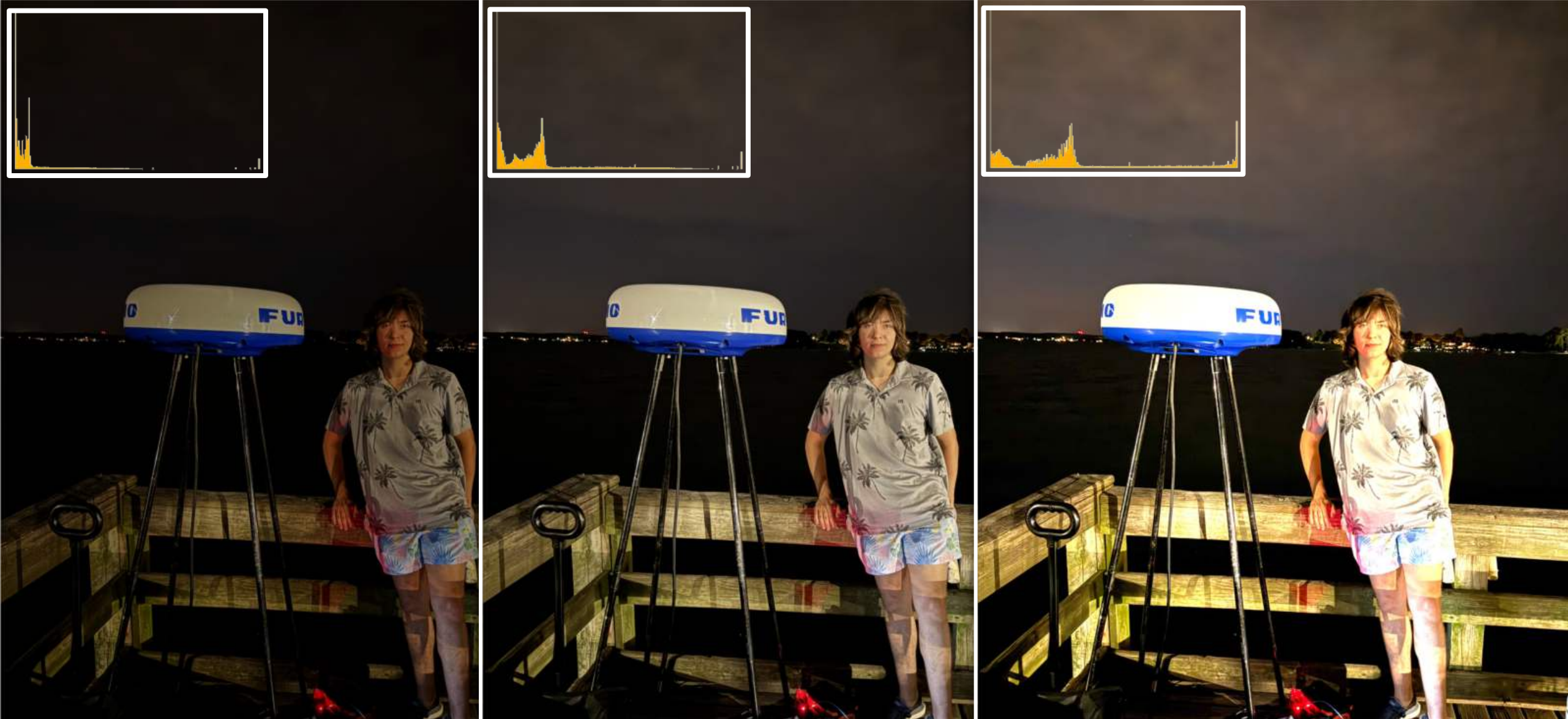
- **Benefits**

- Ideally both faint and strong targets are visible in a single image
 - No special calibration or metadata needed about the gains or exposures
 - The fusion is fast and works out-of-the-box for most scenes

- **Issues**

- Mertens is a heuristic blend. Some physical information about reflectivity might be lost or distorted
 - If the input images are badly misaligned (motion between frames), you get ghosting
 - If some gain settings are very noisy, that noise gets blended into the result





Calibration – Details

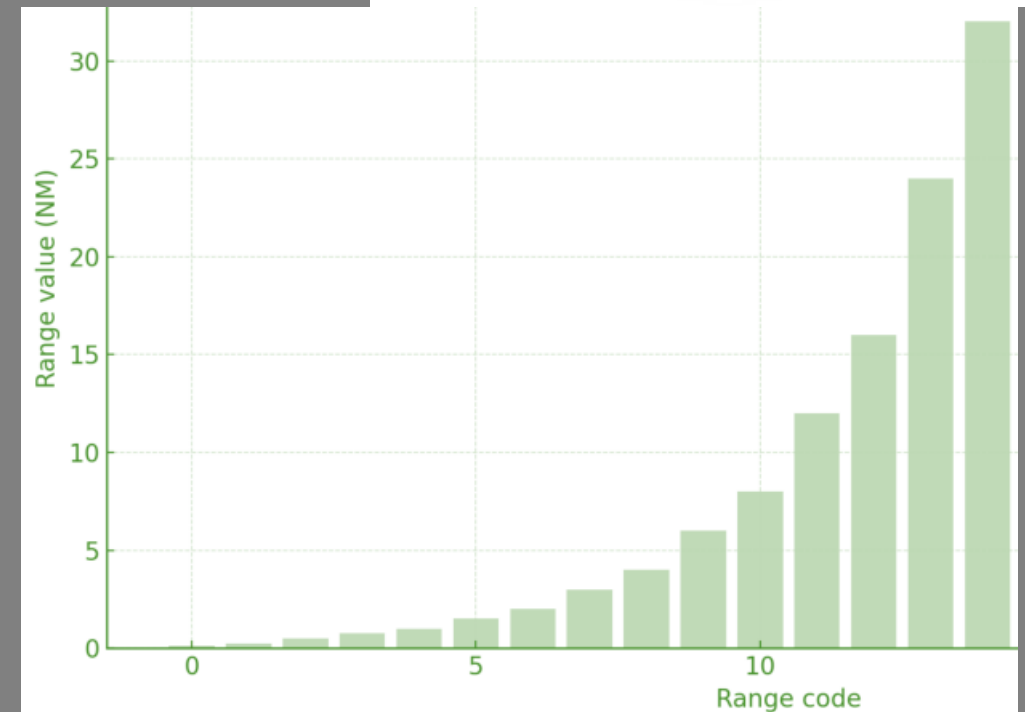
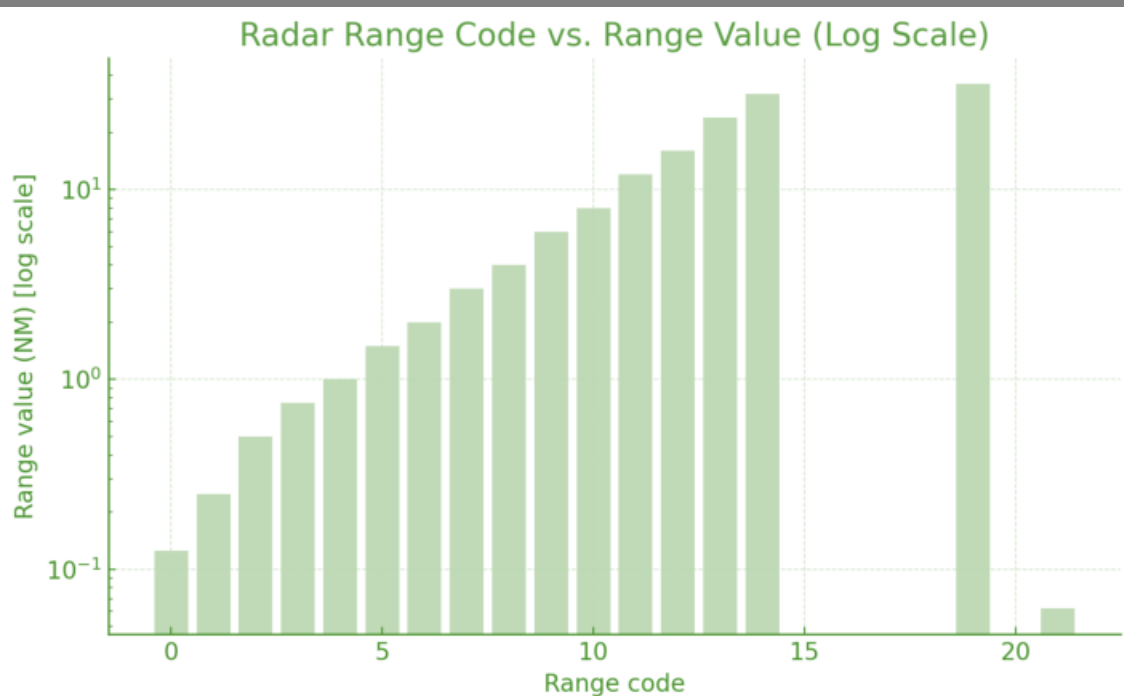
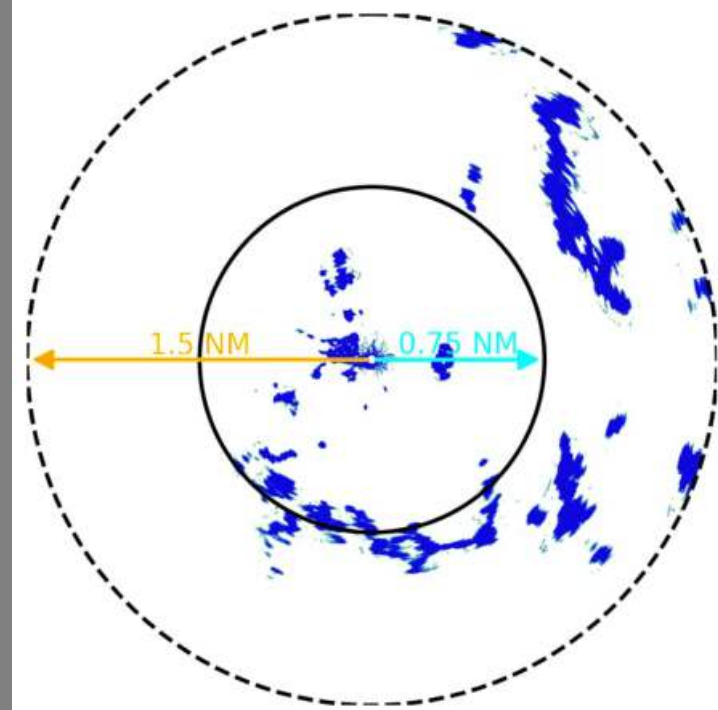
A Summary of the Calibration Techniques, Results, and the path forward

Mini-Calibration

A selection of gain values and ground truth data

What is Range – Furuno

- Range is almost unintuitive
- The range value that is set does not correspond to each ray's length. It corresponds to the halfway point



Radar Waveforms

- **Pulse Radar**

- **Pros:** Simple hardware; high peak power → long range
- **Cons:** Coarse range resolution ($\Delta R \approx c \cdot \tau / 2$); near-field dead zone

- **Pulse-Compression Radar [FURUNO]**

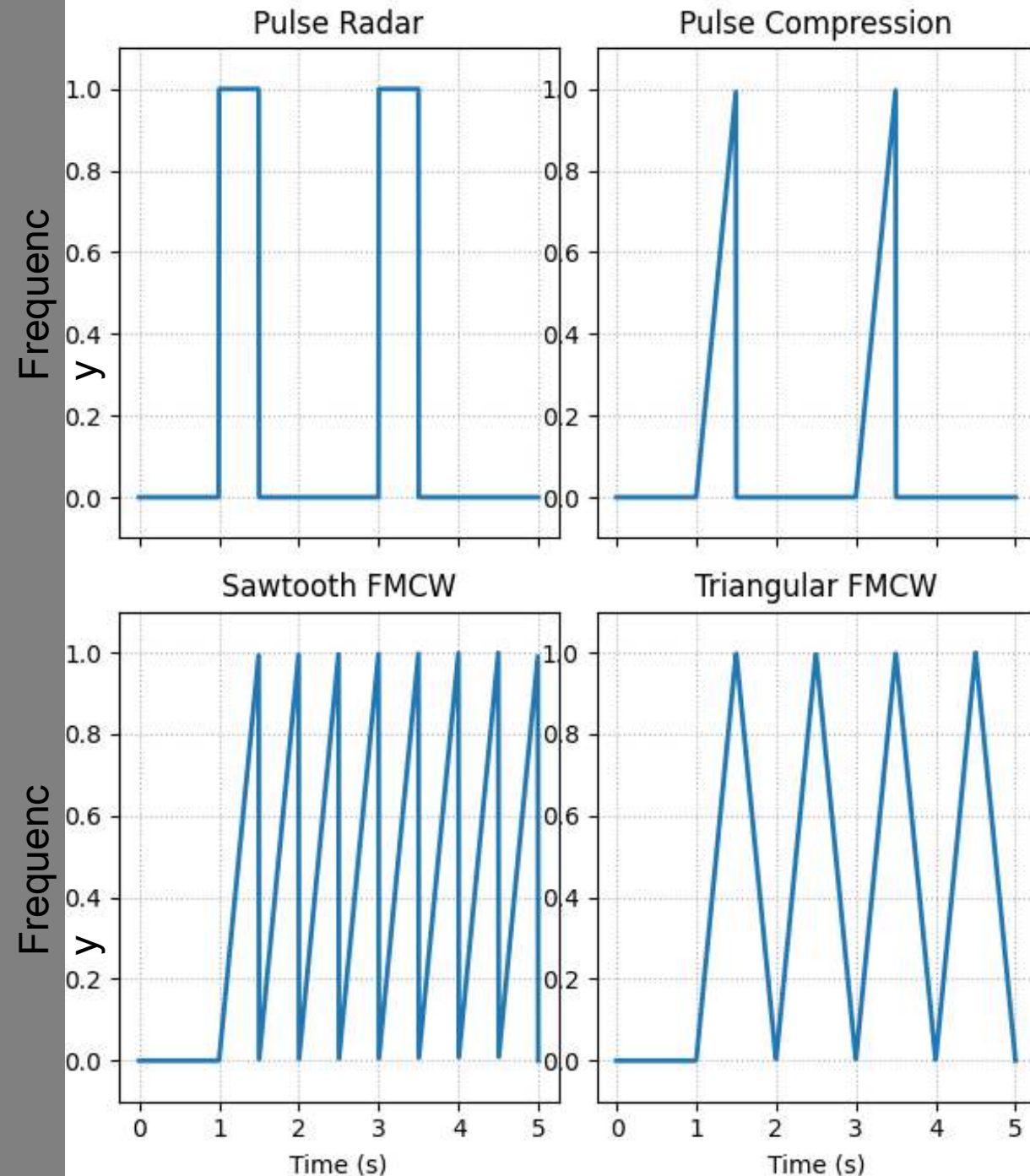
- **Pros:** High resolution & SNR via chirped pulse + matched filter; retains pulse energy without huge peak-power
- **Cons:** Requires complex DSP/ADC hardware; sidelobe suppression needed

- **FMCW Radar (Sawtooth)**

- **Pros:** Continuous transmit → high average power; simultaneous range & velocity extraction via beat frequency
- **Cons:** Lower max range (low peak power); Tx/Rx isolation dead zone; prone to RF interference

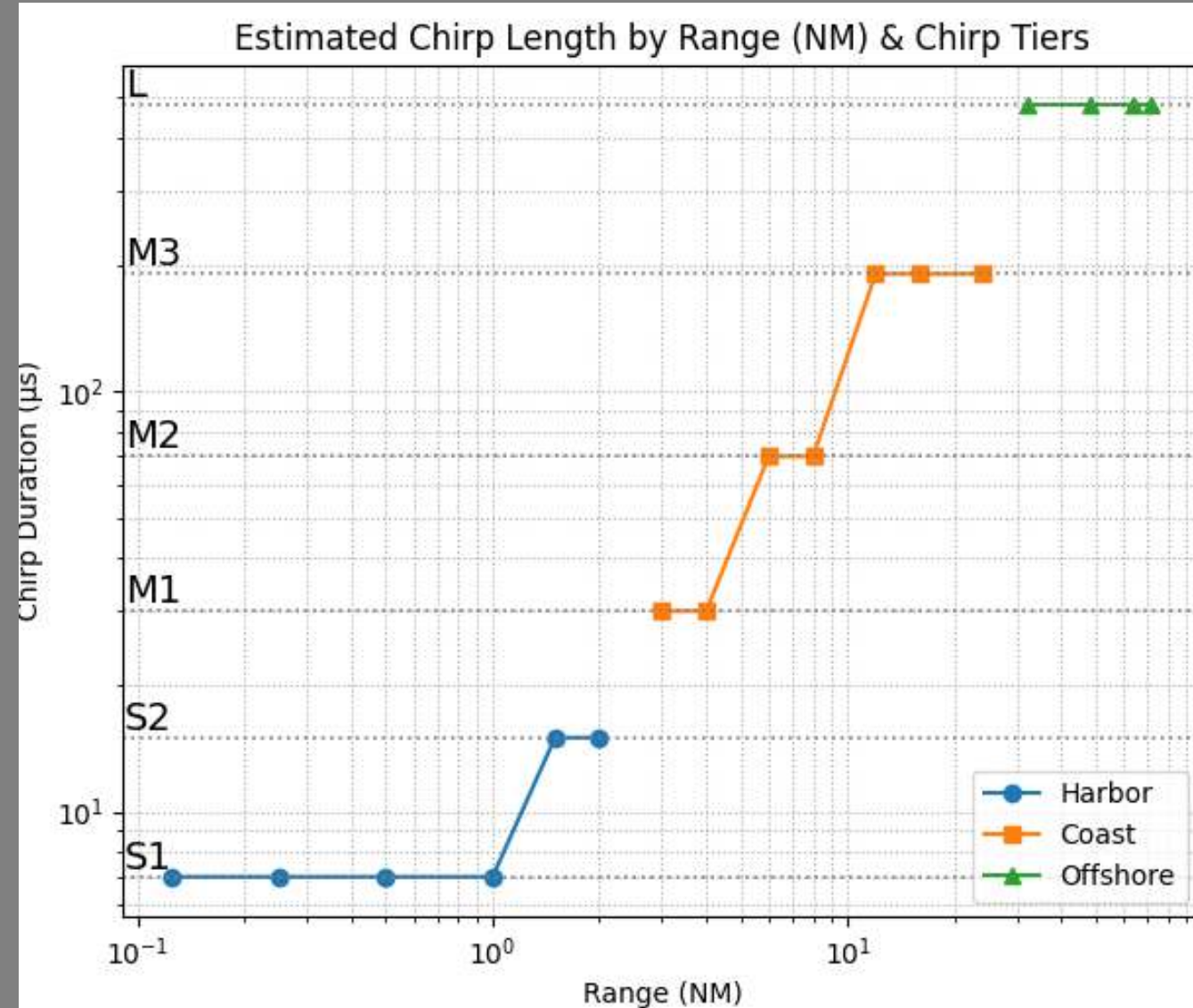
- **Triangular FMCW**

- **Pros:** Up- and down-chirps give two beat tones → unambiguous range & Doppler estimation
- **Cons:** Doubles data volume & processing load; more complex waveform generation & timing



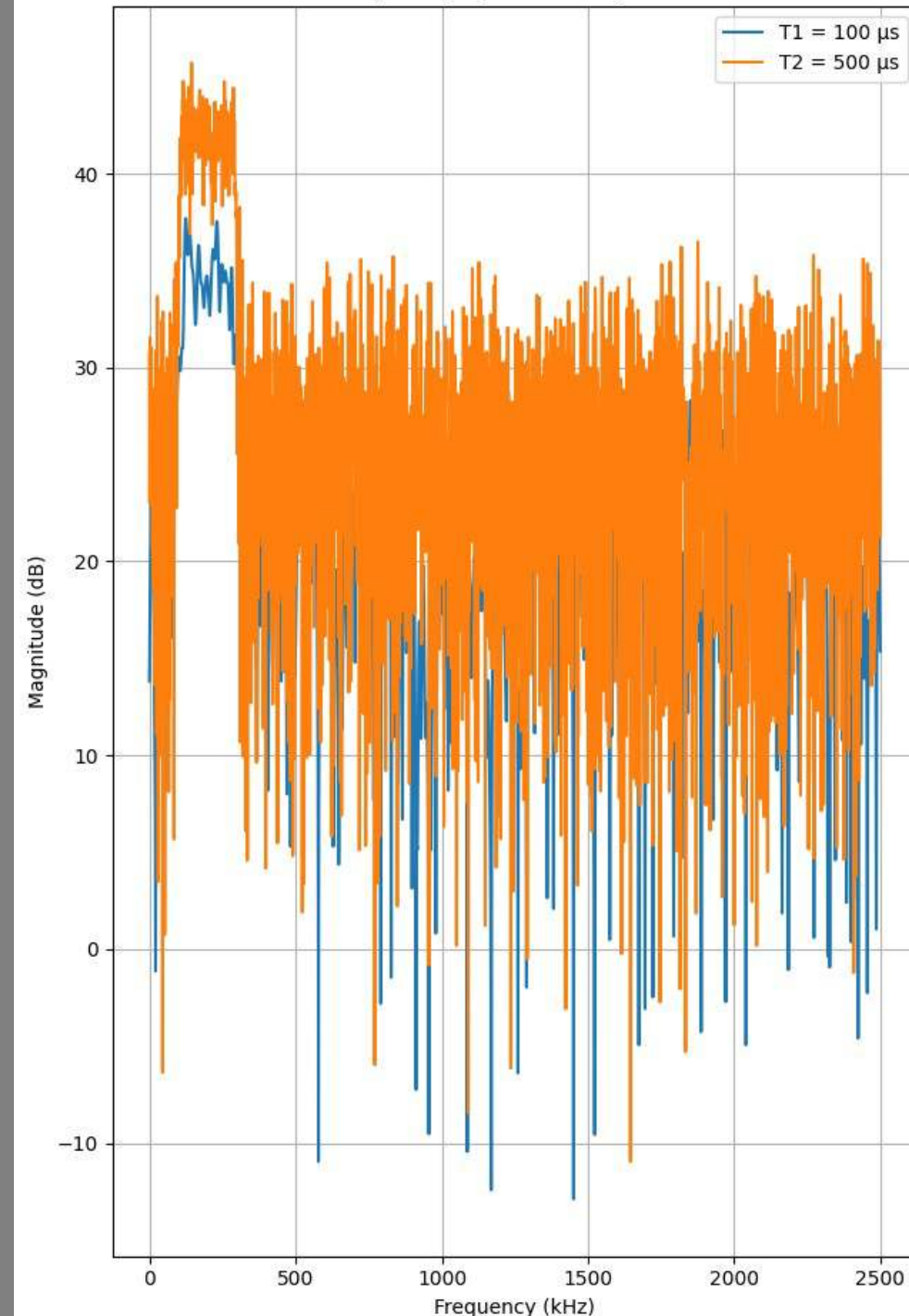
Range and Pulse Length/Width – Furuno

- Furuno has 6 different chirp lengths
 - 2 Short Chirps
 - 3 Medium Chirps
 - 1 Long Chirp
- Testing completed during 'calibration' only uses S1 and S2 chirps
- S2 Chirp appears to more reliably detect buoys at varying gains

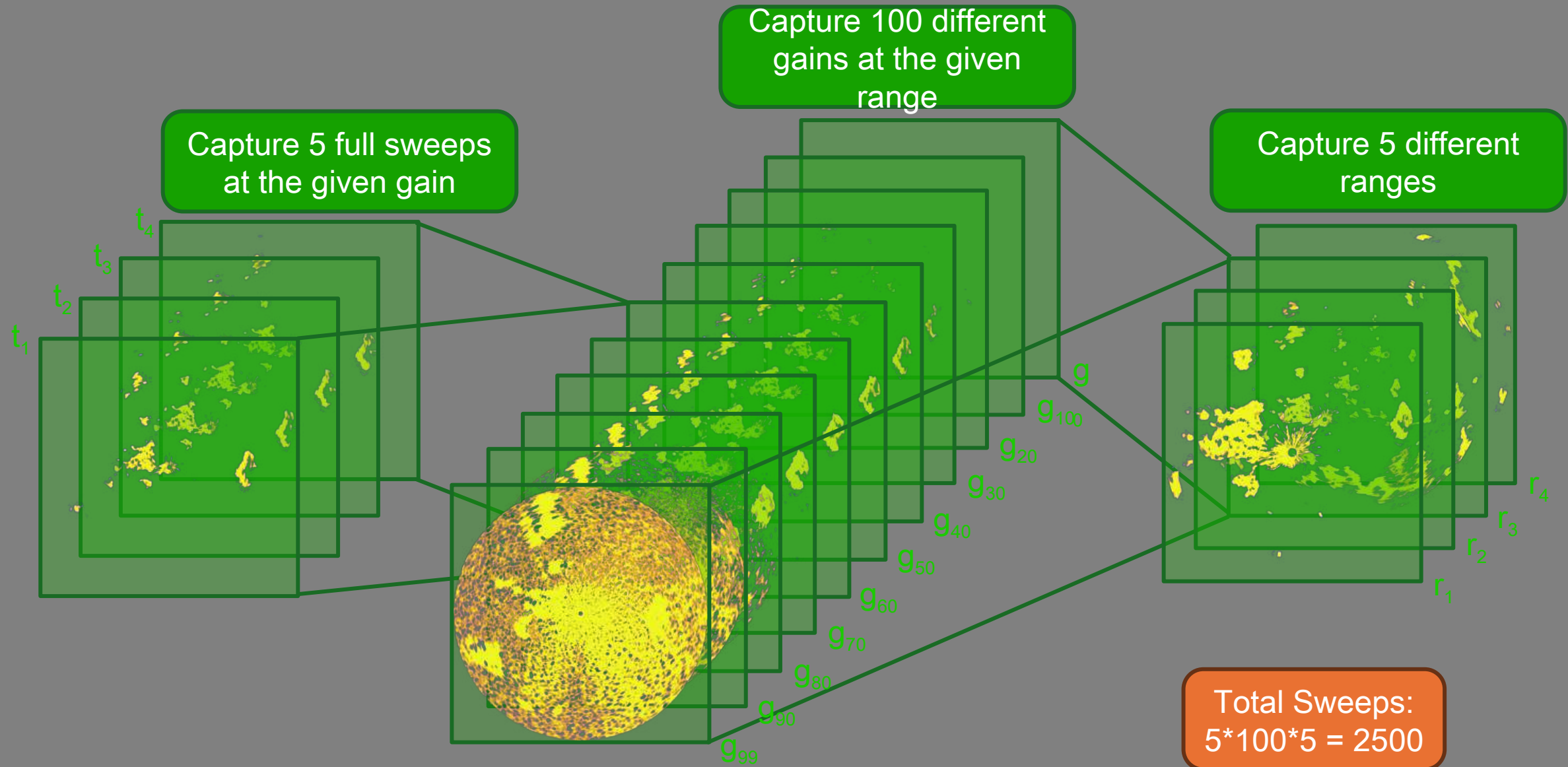


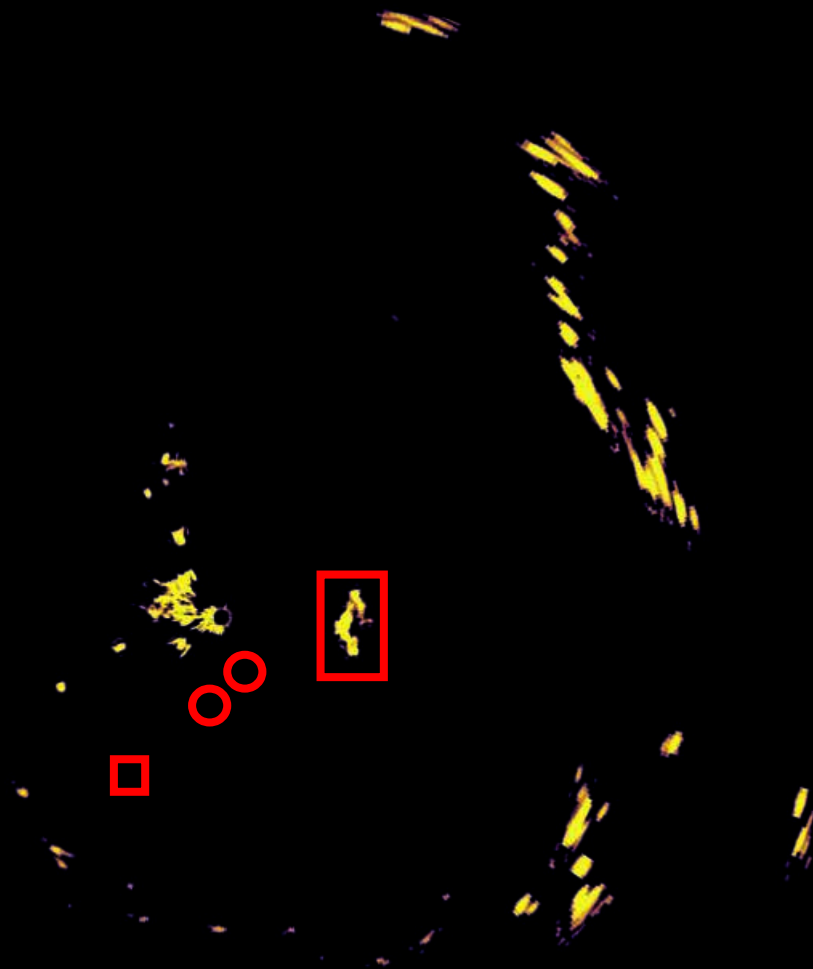
Pulse Width & Noise

- **Higher Matched-Filter Gain**
 - Gain $\propto$ Time–Bandwidth Product ($TBP = T \times B$) $\rightarrow$ higher output SNR & better detection reliability
- **Improved Detection Probability**
 - Greater SNR lowers the threshold for a given P_d/P_{fa}
- **Range Resolution Unchanged**
 - $\Delta R = c/(2B)$, depends only on bandwidth B , not on pulse length T
- **Noise Handling**
 - Concentrates in-band signal energy into a narrow spike; does not remove out-of-band noise
- **Trade-offs**
 - Longer T demands more ADC sampling, memory & DSP for matched filtering
 - Excessive T may cause Doppler smear or range ambiguities unless PRF/waveform is adjusted

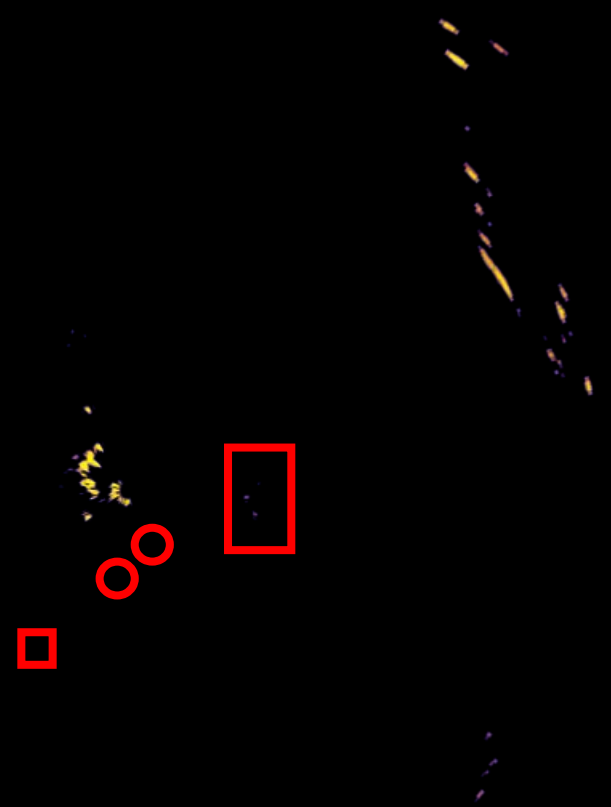


'Calibration' Sequence

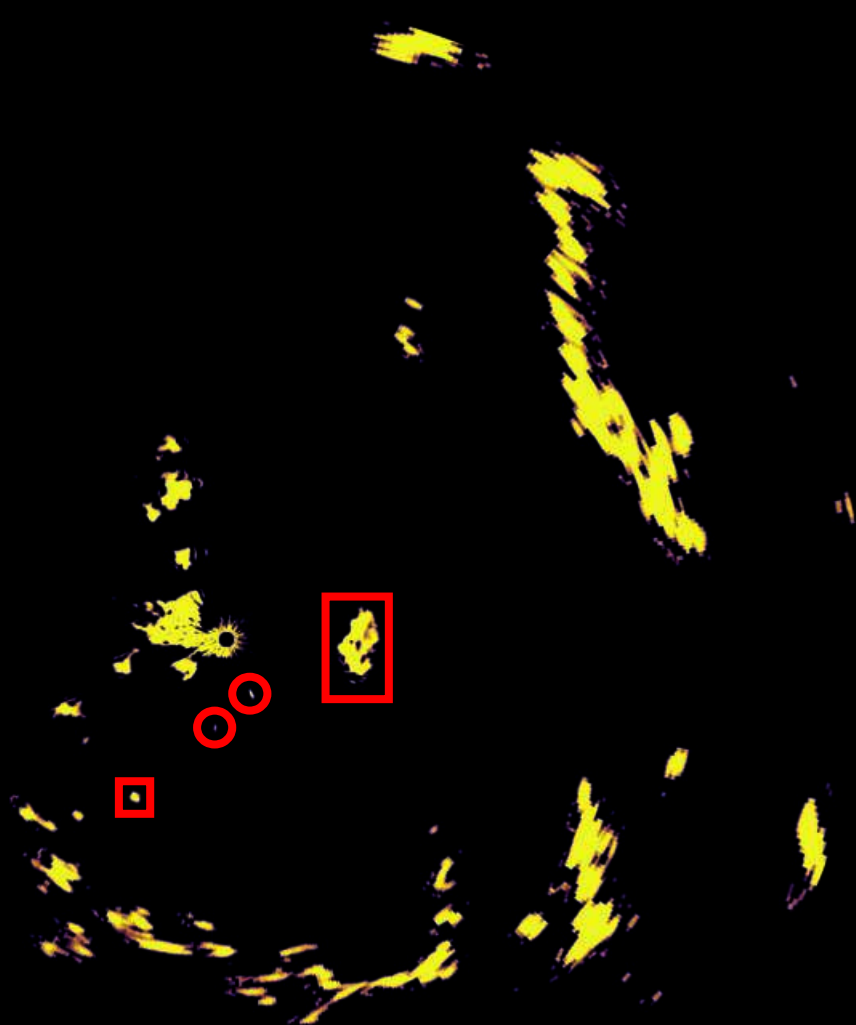




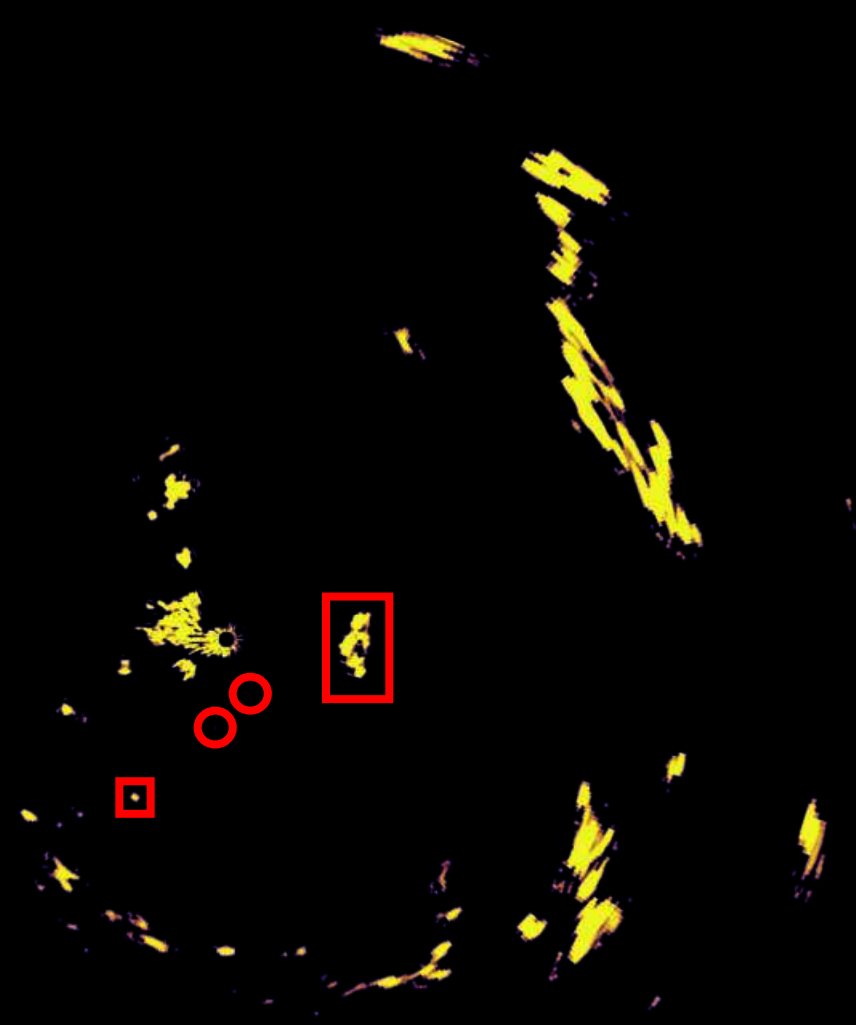
Gain
20



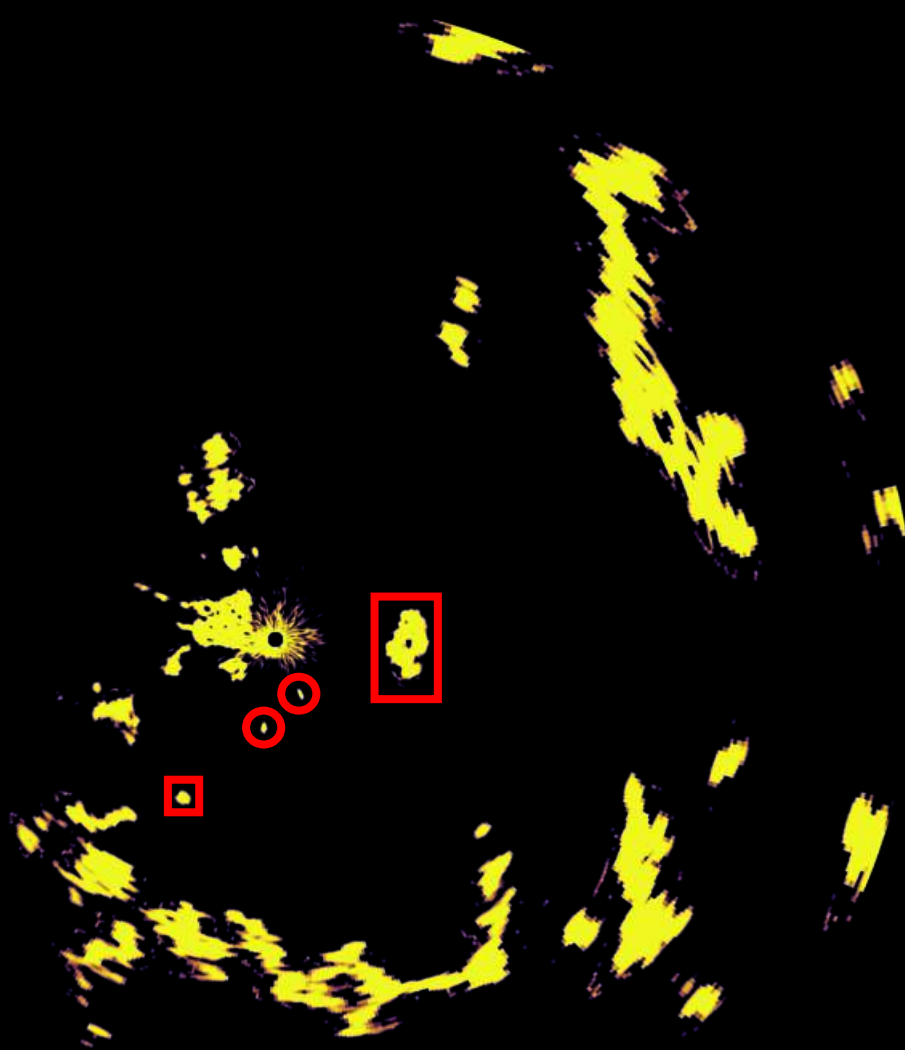
Gain
10



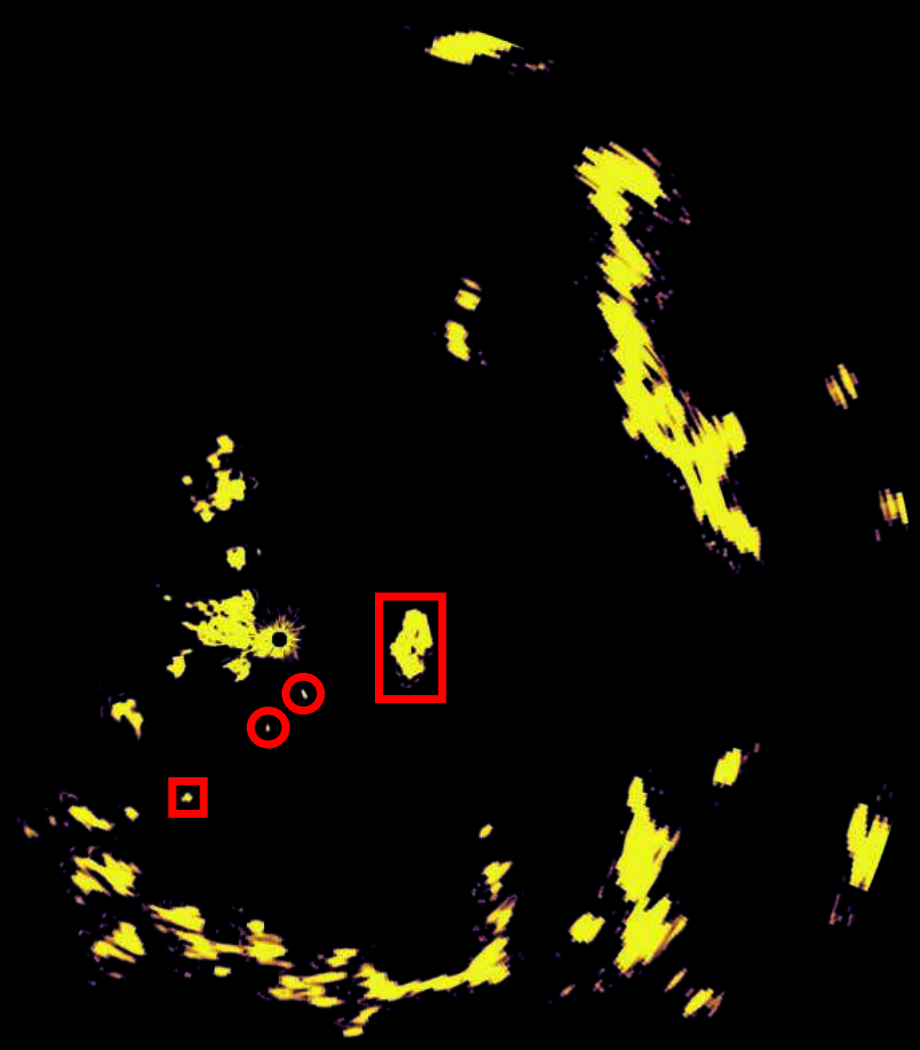
Gain
40



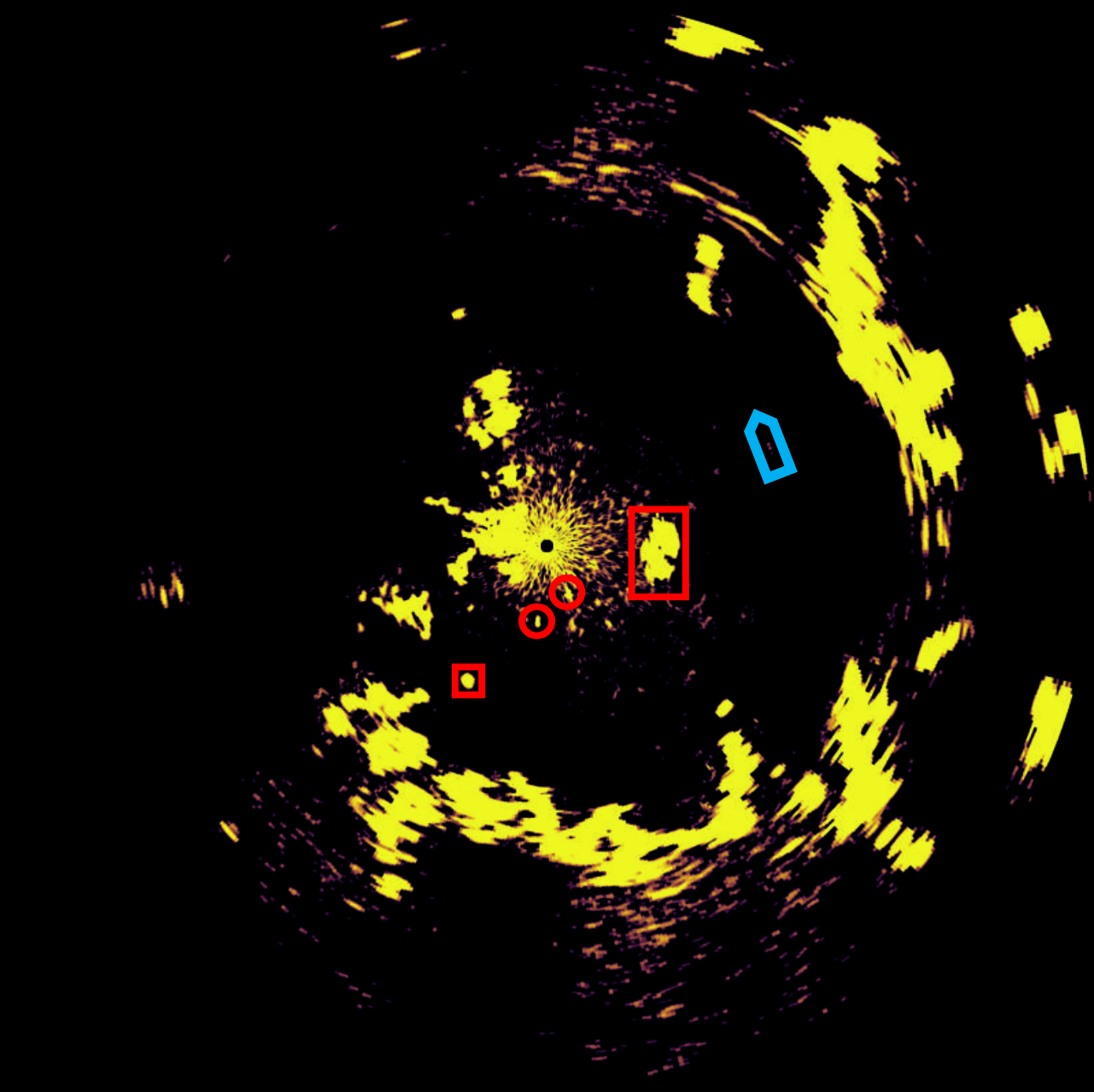
Gain
30



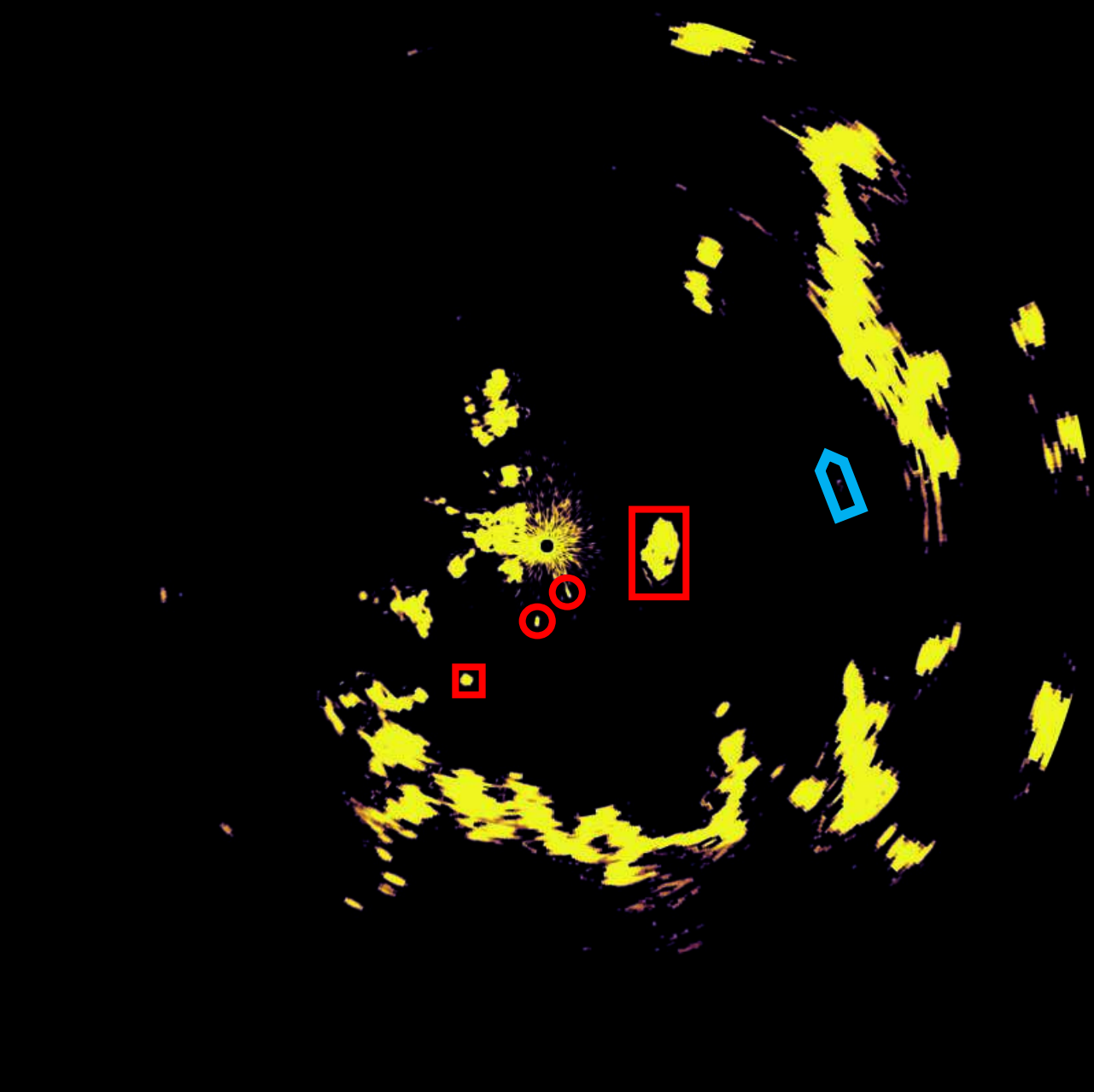
Gain
60



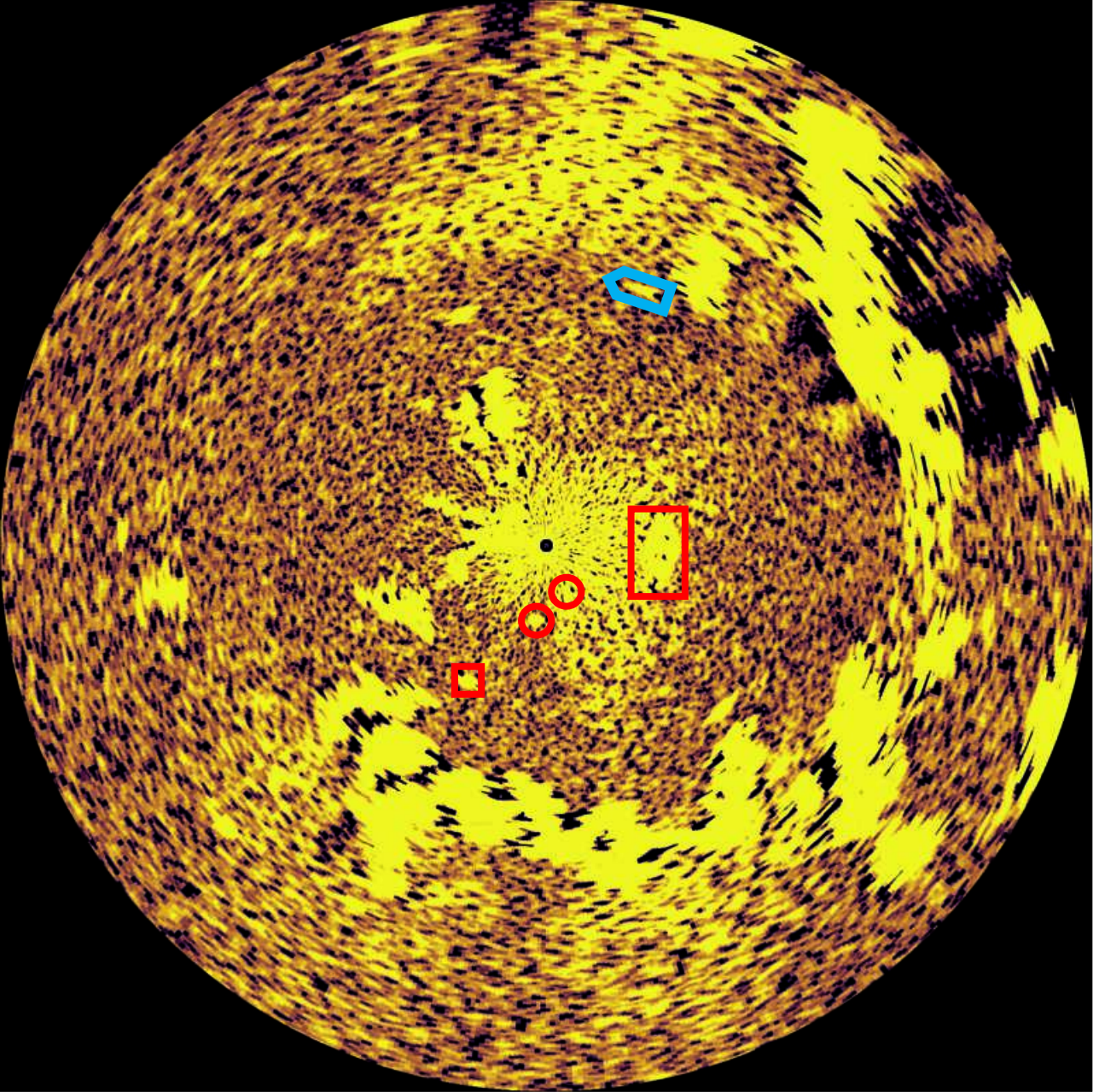
Gain
50



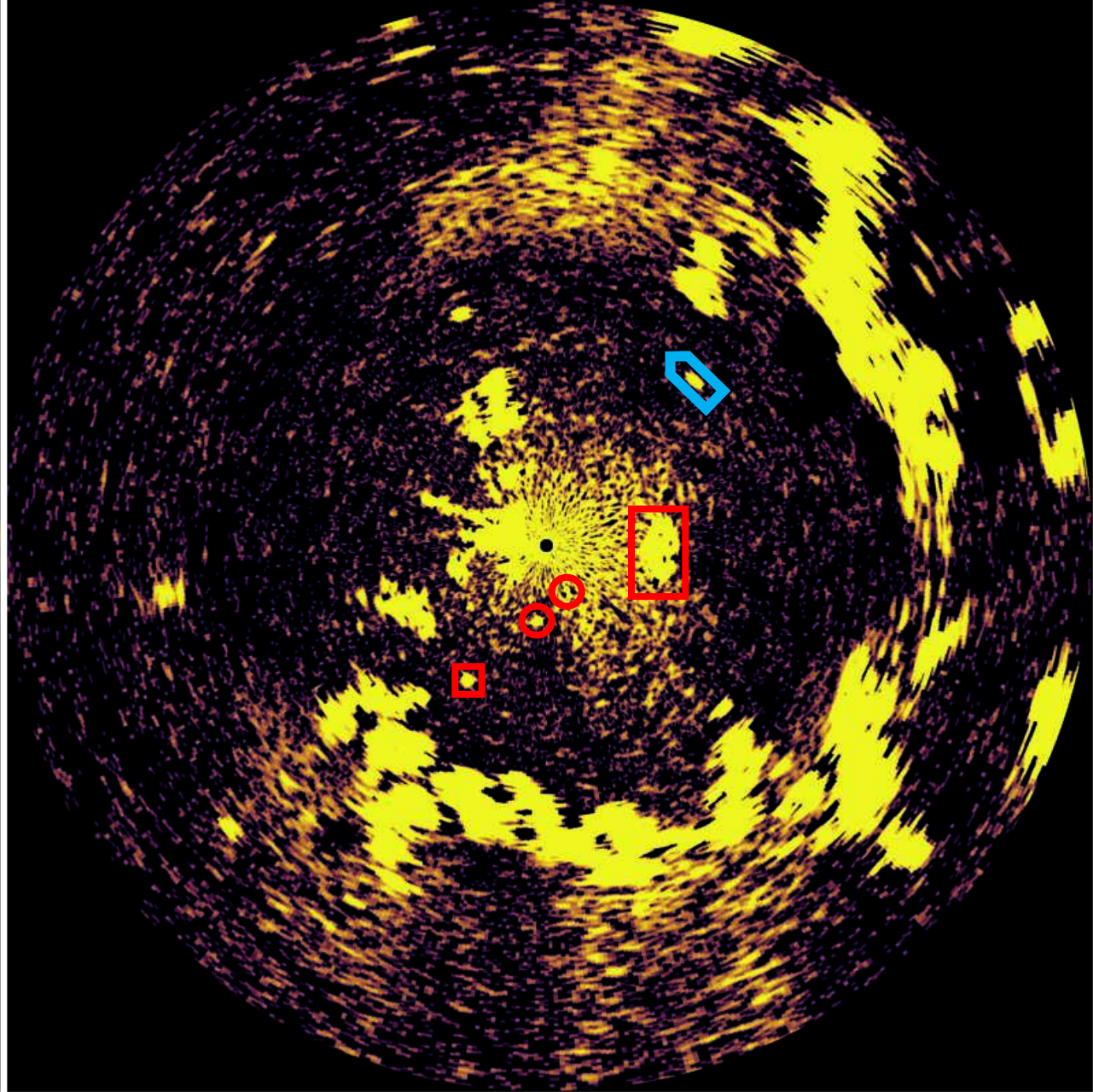
Gain
80



Gain
70



Gain
99



Gain
90

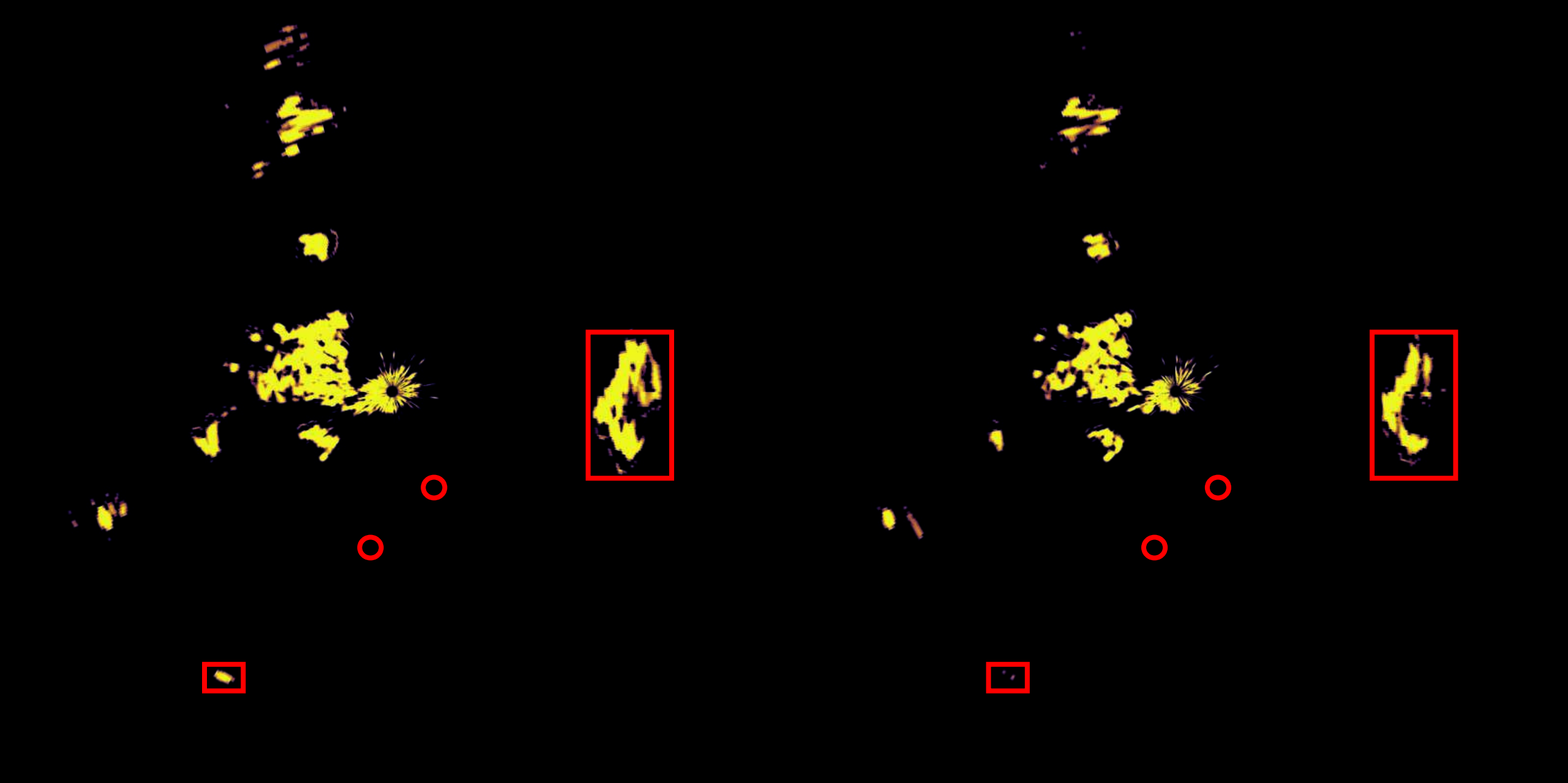
Summary for Range 3 (0.75 NM)

- Gain 30, most land masses are detected while no small objects are detected
- Gain 70, Faint detection of far, moving objects
- Gain 40, good detection of buoys
- Gain 90, Good detection of far, moving object, albeit high noise content
- Gain 99/100, Everything detected, only useful for identify large land masses/objects and rain/waves



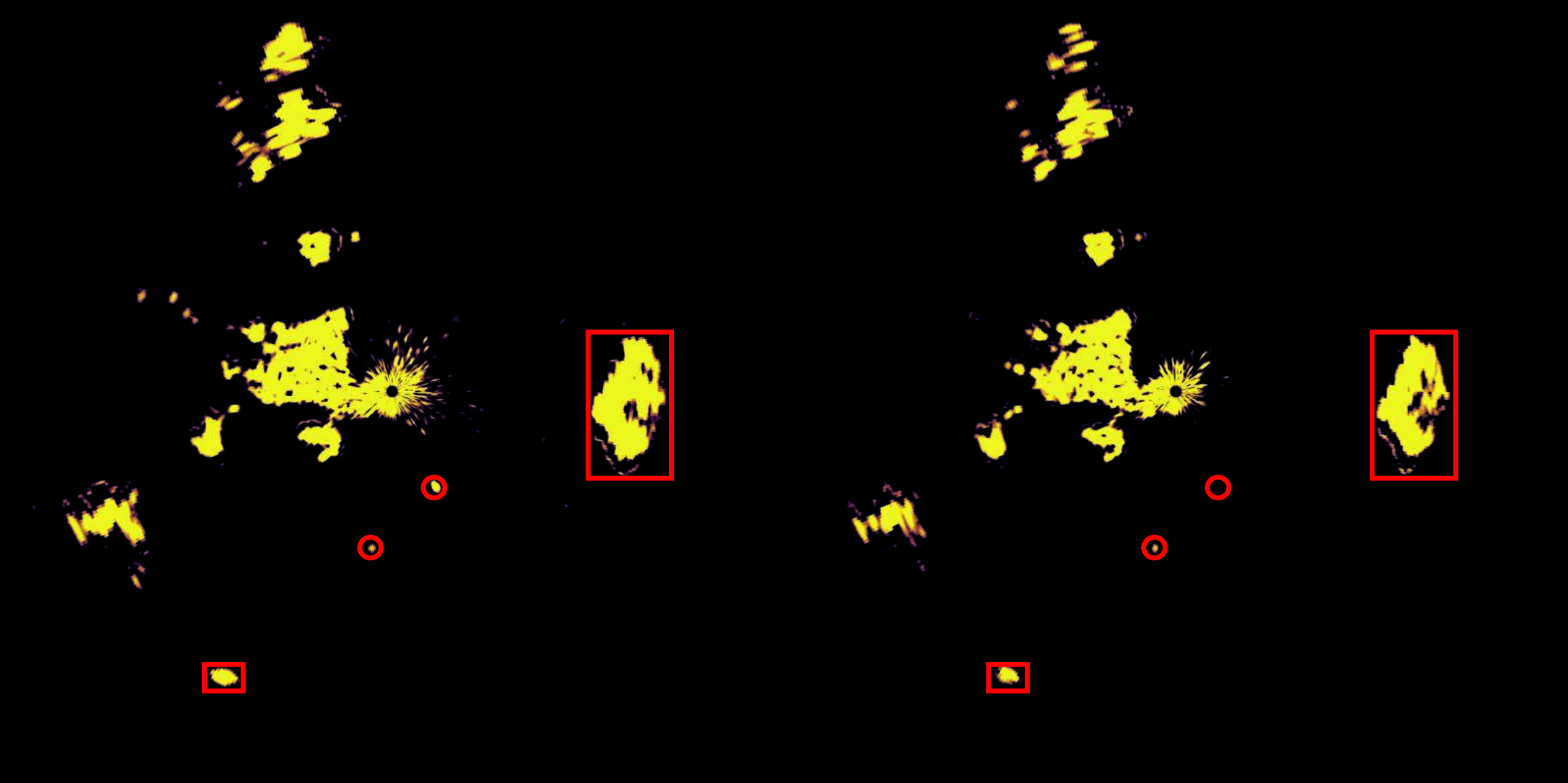
Gain
20

Gain
10



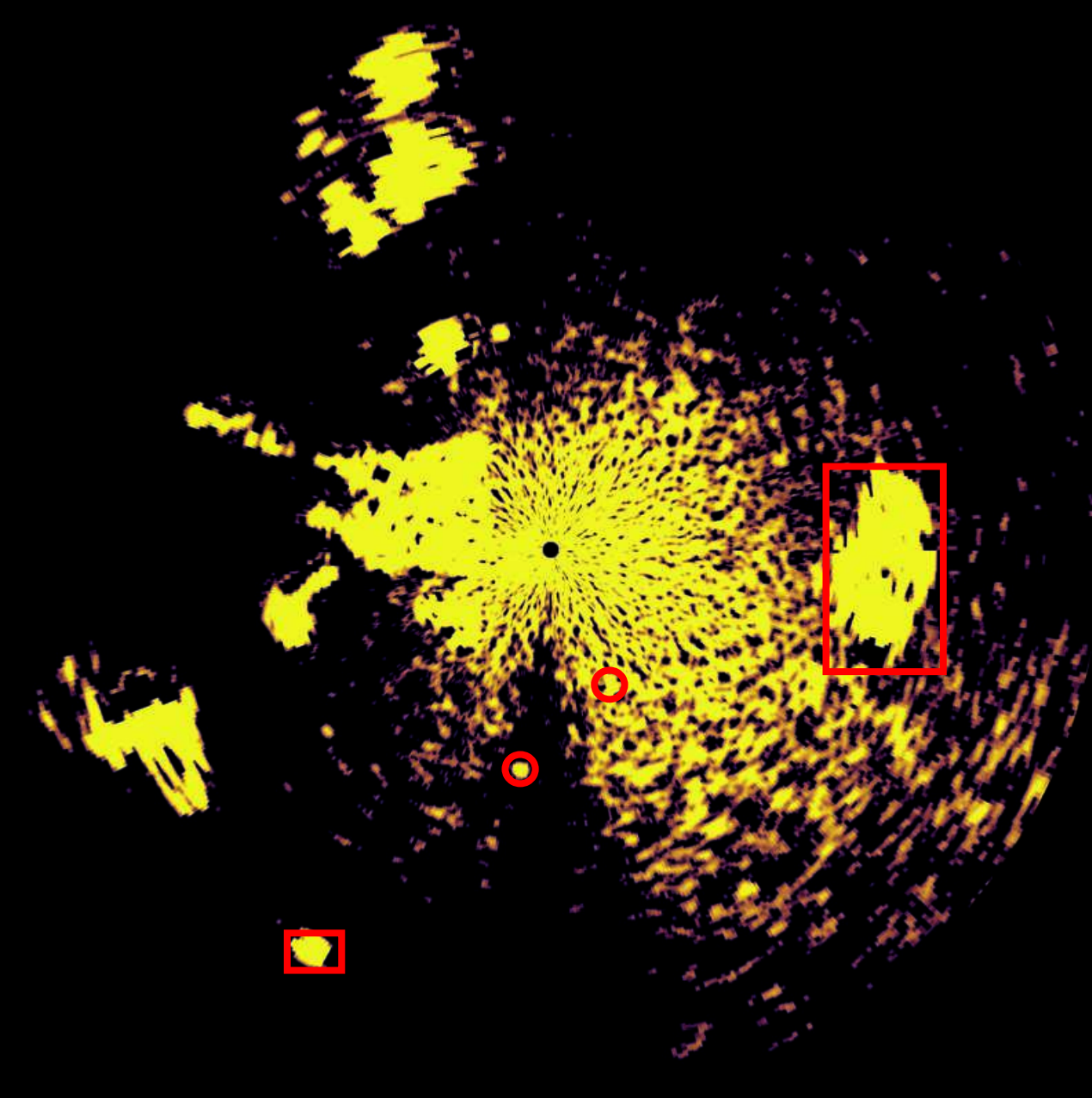
Gain
40

Gain
30

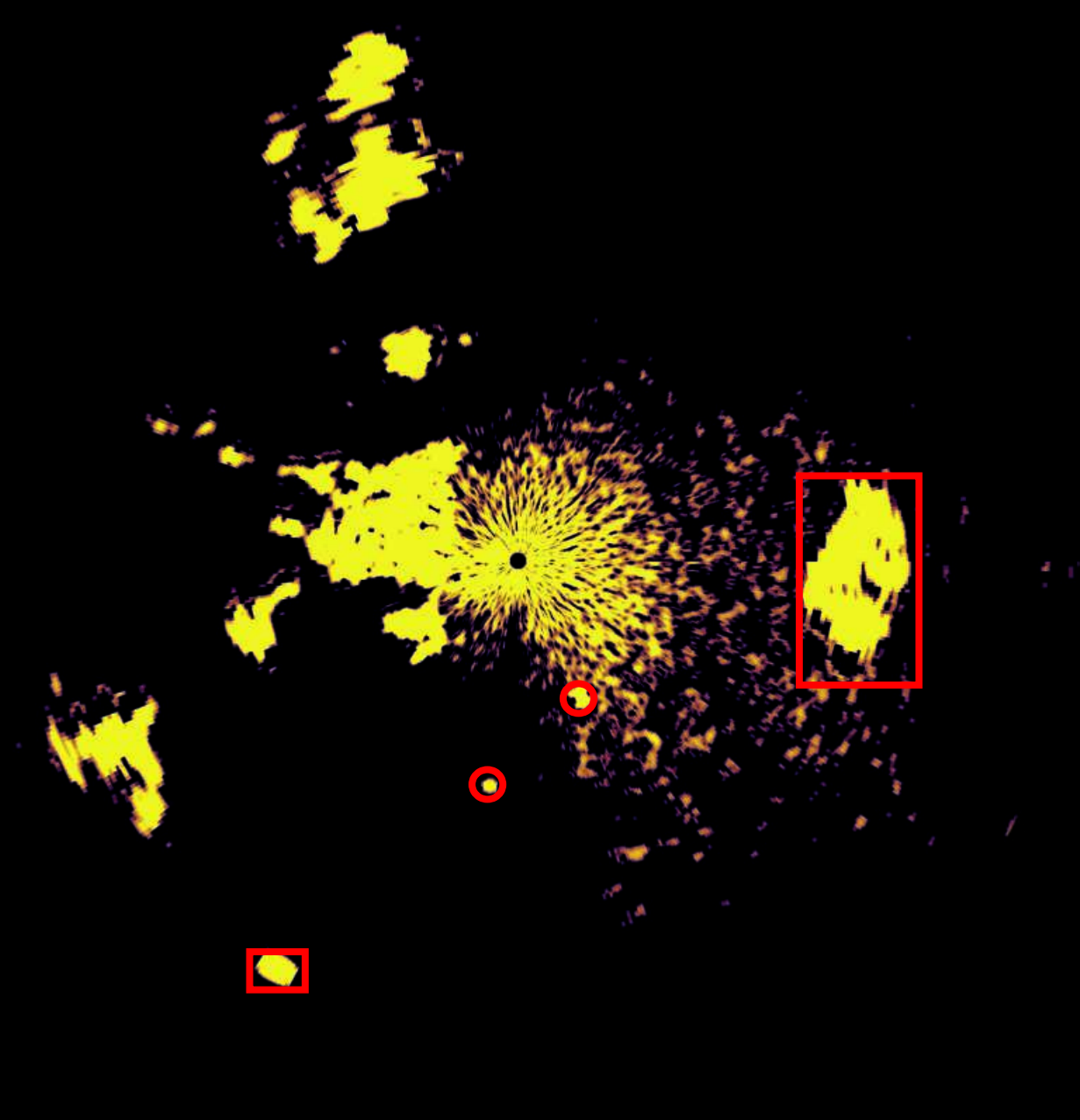


Gain
60

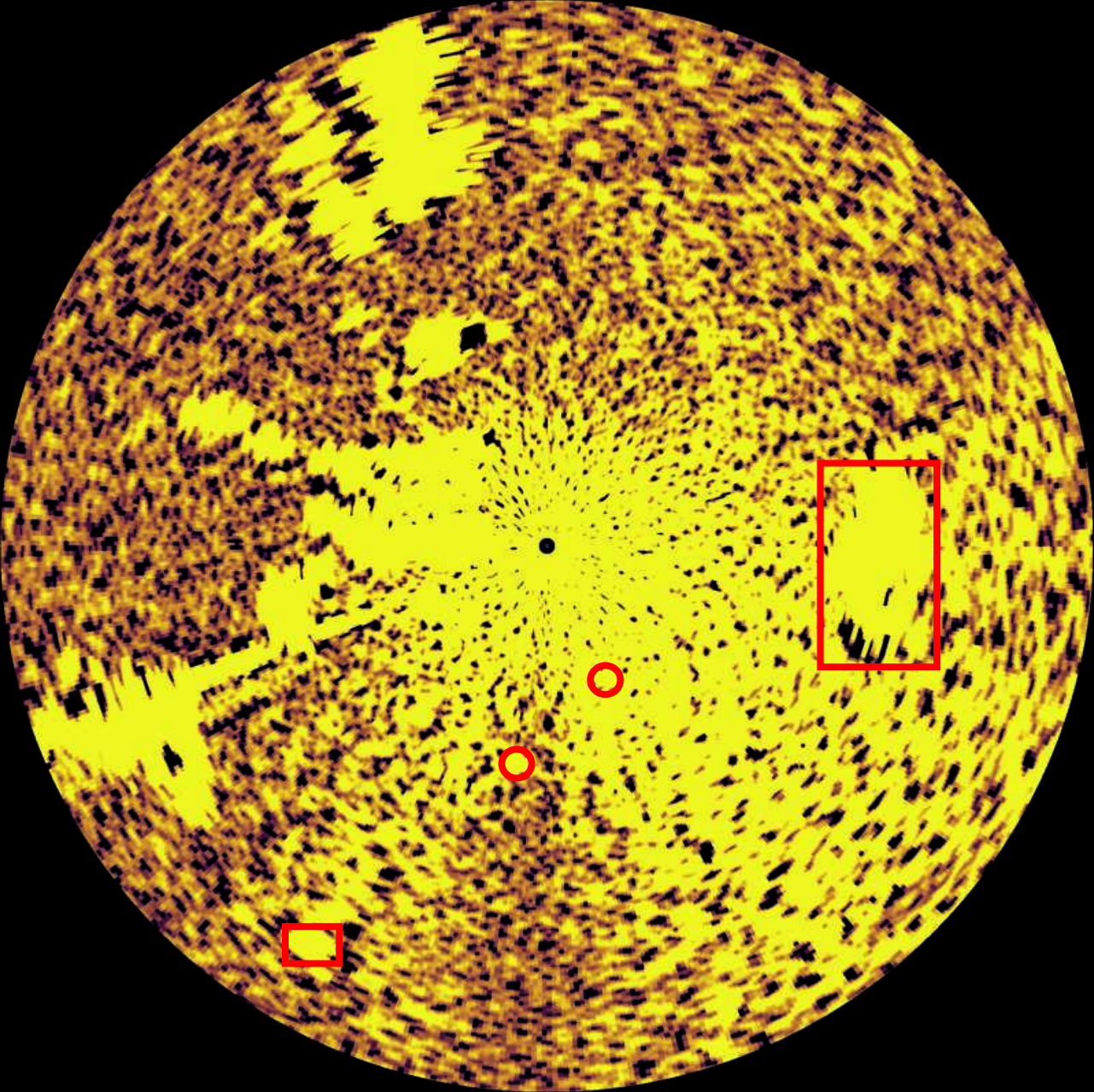
Gain
50



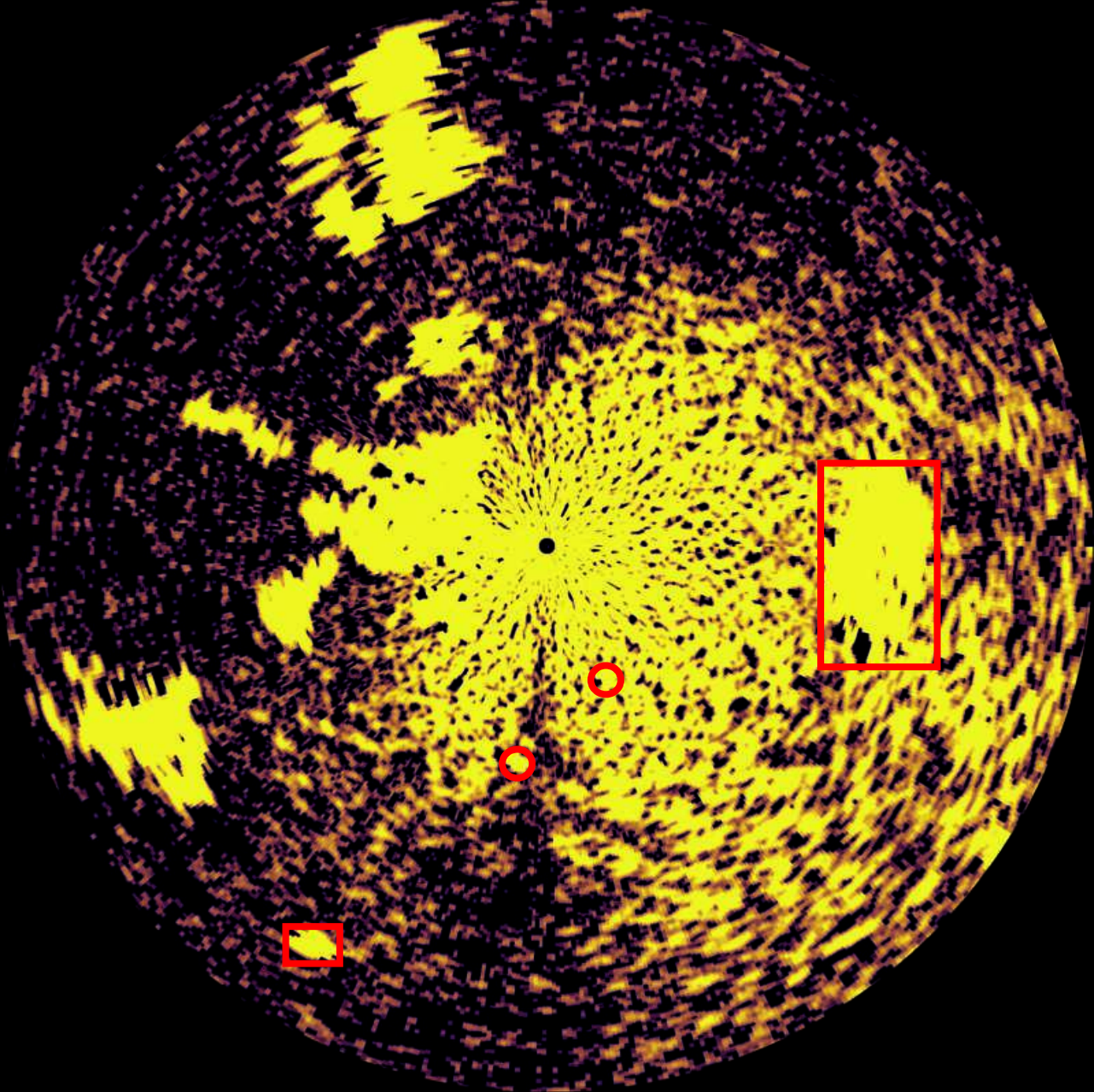
Gain
80



Gain
70



Gain
99



Gain
90

Summary for Range 1 (0.25 NM)

- Gain 40, most land masses are detected while no small objects are detected
- Gain 60, Good detection of buoys/stationary objects
- Gain 70, may be able to parse low-RCS objects.
- Gain 80, hard to parse low-RCS objects
- Gain 99/100, Everything detected, not useful.

The background is a vibrant blue with a repeating pattern of colorful speech bubbles. Each bubble is a different color (red, yellow, pink, grey, olive green, purple) and contains a large, dark blue question mark. The bubbles are scattered across the entire frame, creating a sense of inquiry and communication.

Thank you.

Any Questions?

Code is attached here.

N2V Code(AI based
declutter)

All other code

RPCA library